Constrained Energy Minimization Anomaly Detection for Hyperspectral Imagery via Dummy Variable Trick

Chein-I Chang, *Life Fellow, IEEE*

***Abstract*— CEM has shown great success in subpixel target detection. This article develops a DVT to extend CEM to CEM-AD and shows that CEM-AD also enjoys the same success in anomaly detection (AD). Its idea converts a known specific target signature d imposed on CEM into an unknown specific target signature to develop an SBR-CEM as a UST-CEM which serves as a liaison to derive the desired CEM-AD without prior knowledge of d. Surprisingly, the derived CEM-AD turns out to be a sample correlation matrix R-based AD, R-AD in correspondence to the well-known sample covariance matrix K-based AD developed by Reed-Xiaoli, RX-AD. To further improve CEM-AD, an LRaSMD model introduced by GoDec and its SC are further incorporated into CEM-AD where two new versions of SC, FSC and VSC, are particularly designed to enhance AD. Finally, to effectively evaluate AD performance, recently developed 3-D ROC curve-derived detection measures are used for comparative studies and analyses.**

***Index Terms*— Anomaly detection (AD), CEM, CEM-AD, DVT, UST-CEM.**

LIST OF ACRONYMS

AD	Anomaly detection.
AUC	Area under an ROC curve.
ATGP	Automatic target generation process.
BKG	Background.
BS	Background suppression.
BRP	Bilateral random projection.
CDA	Component decomposition analysis.
CEM	Constrained energy minimization.
CEM-AD	CEM anomaly detector.
CRD	Collaborative representation detector.
DVT	Dummy variable trick.
FSC	Fixed sparsity cardinality.
FIR	Finite impulse response.
GoDec	Go decomposition.
GLRT	Generalized likelihood ratio test.
HAD	Hyperspectral anomaly detection.
HTD	Hyperspectral target detection.
ICs	Independent components.
LRaSMD	Low rank and sparse matrix decomposition.
MX-SVD	Minimax singular value decomposition.
ODP	Overall detection probability.
OSP	Orthogonal subspace projection.
OSP-AD	OSP anomaly detector.
PC	Principal component.
RX-AD	Reed-Xiaoli anomaly detector.
ROC	Receiver operating characteristic.
RPCA	Robust principal component analysis.
RSL	Robust subspace learning.
SC	Sparsity cardinality.
SBR	Signal-to-BKG ratio.
SNPR	Signal-to-noise probability ratio.
SNR	Signal-to-noise ratio.
SVM	Support vector machine.
TD	Target detectability.
UST	Unknown specified target.
VD	Virtual dimensionality.
VSC	Variable SC.

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The author is with the Center for Hyperspectral Imaging in Remote Sensing (CHIRS), Information and Technology College, Dalian Maritime University, Dalian 116026, China, also with the Remote Sensing Signal and Image Processing Laboratory, Department of Computer Science and Electrical Engineering, University of Maryland, Baltimore, MD 21250 USA, and also with the Department of Computer Science and Information Management, Providence University, Taichung 02912, Taiwan (e-mail: cchang@umbc.edu).

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I. INTRODUCTION

HAD has received considerable interest lately. A detailed review on anomaly detection (AD) can be found in [1]. It is quite different from HTD [2] in the sense of their detection modes, passive TD for HAD and active TD for HTD according to taxonomy in [3, chapter 5]. As for HTD, the targets to be detected are assumed to be known by *a priori* or *a posteriori* knowledge compared to HAD which does not have any target knowledge at all. As a result, the detectors to be designed are generally quite different. One cannot be applied to another. This is because how can we make the requirement of the known target knowledge used by HTD go away for HAD?

In a classic approach, the binary composite hypothesis testing was used to derive the GLRT, which was used by Reed and Xiaoli to derive an AD, called RX-AD [4]. Since GLRT requires target knowledge for the hypothesis testing problem, it further assumed that there was a secondary dataset to be used to specify the target shape as a known signature so that the unknown target amplitude can be estimated by maximum likelihood estimator to derive RX-AD [5], [6]. Chang [1] took a completely different approach from GLRT without assuming such an auxiliary dataset. Instead, it introduced a UST, $\mathbf{t}^{\mathrm{UST}}$, as a dummy variable where $\mathbf{t}^{\mathrm{UST}}$ is the target assumed to be "*specified*" implicitly but "*unknown*" in nature without prior knowledge. Most importantly, $\mathbf{t}^{\mathrm{UST}}$ is not

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probability-distributed but rather a data sample which can be used as a liaison to make a transition from known specified target to UST.

In HTD, two widely used subpixel target detectors are OSP-TD developed in [7] and [8] and the CEM detector developed in [8]–[10]. Both require the prior knowledge of the target to be detected. Recently, Chang *et al.* [11] utilized an unsupervised OSP-TD, called ATGP [12], as a liaison to convert OSP-TD into an OSP-AD without specifying targets to be detected. Since there is no counterpart of ATGP found in CEM, it seems not feasible to make CEM an anomaly detector because CEM imposes a constraint on its detector structure, while OSP does not. This article shows otherwise. It takes advantage of the concept of UST, $\mathbf{t}^{\mathrm{UST}}$, to make OSP-TD and CEM anomaly detectors. In order to make it work, we introduce a new version of CEM, called SBR-CEM, which replaces the desired known target signature $\mathbf{d}$ and the unity constraint, respectively, with $\mathbf{t}^{\mathrm{UST}}$ and its SBR as a constraint imposed on CEM. Such use of $\mathbf{t}^{\mathrm{UST}}$ is referred to as DVT because $\mathbf{t}^{\mathrm{UST}}$ is a dummy variable and will vanish eventually. As a matter of fact, by virtue of SBR-CEM, CEM can be extended to CEM which imposes a constraint of a UST on CEM by UST-CEM and then further extended to CEM-AD with the UST-CEM vanished after all. More specifically, we reinterpret the minimum variance derived from CEM for $\mathbf{t}^{\mathrm{UST}}$ as its SBR, denoted by $\mathrm{SNR}(\mathbf{t}^{\mathrm{UST}})$, so that this $\mathrm{SNR}(\mathbf{t}^{\mathrm{UST}})$ can be used as a constraint imposed on UST-CEM instead of unity imposed on CEM. So, maximizing $\mathrm{SNR}(\mathbf{t}^{\mathrm{UST}})$, i.e., $\max_{\mathbf{t}^{\mathrm{UST}}}\mathrm{SNR}(\mathbf{t}^{\mathrm{UST}})$ yields CEM-AD where $\mathbf{t}^{\mathrm{UST}}$ is eventually disappeared. This $\mathbf{t}^{\mathrm{UST}}$ plays a similar role that the secondary dataset plays in GLRT in [4] and [6] used for maximum likelihood estimation of the unknown target amplitude and is used to bridge the target knowledge discrepancy between HTD in [2] and HAD in [1] during target knowledge transition. Its idea is similar to the kernel trick [13] used by SVM to calculate inner product of a kernel matrix with no need of specifying the used kernel function ϕ which is similar to an unknown but specified target $\mathbf{t}^{\mathrm{UST}}$ as a dummy target.

Interestingly, using a similar approach via DVT, OSP-TD can be extended to a UST-OSP and then extended to OSP-AD in the same manner that UST-CEM is used to extend CEM-AD. Surprisingly, such OSP-AD turns out to be exactly the same as OSP-AD derived in [11] using ATGP.

However, there is a key and prominent difference between CEM-AD and OSP-AD in the sense of how they deal with the BKG issue. That is, CEM-AD suppresses BKG by inverting the sample correlation matrix $\mathbf{R}$, i.e., $\mathbf{R}^{-1}$, as opposed to OSP-AD which utilizes an OSP projection matrix to null the BKG. Accordingly, from their functionalities, CEM-AD and OSP-AD are derived from completely different rationales. Nevertheless, they both share the same idea that is how to appropriately represent BKG prior to its suppression by CEM-AD or BKG removal by OSP-AD.

Once BKG is taken care of by $\mathbf{R}^{-1}$ or OSP, a follow-up issue is how to enhance AD. Several approaches have been investigated in the past. Of particular interest is the LRaSMD model by GoDec proposed in [14] which decomposes a data space $\mathbf{X}$ into three component spaces as $\mathbf{X} = \mathbf{L} + \mathbf{S} + \mathbf{N}$ where $\mathbf{L}$, $\mathbf{S}$, and $\mathbf{N}$ represent low rank space, sparse space, and noise spaces, respectively. In particular, GoDec introduced the concept of SC which can extract anomalies from the sparse space $\mathbf{S}$. Since CEM-AD performs BKG suppression (BS) by performing $\mathbf{R}^{-1}$ to suppress BKG and detecting targets by a matched filter using the desired target signature as the matched signal, we can operate CEM-AD in two separate and individual spaces to suppress BKG in one space, $\mathbf{L}$ while extracting anomalies in another separate space, SC-constrained $\mathbf{S}$ instead of the same data space as CEM does. This LRaSMD model provides a perfect fit for CEM-AD. Specifically, SC is further generalized to two versions, FSC and VSC imposed on the sparse space S to enhance anomaly extraction.

To evaluate BKG effects on AD, the commonly used ROC has been shown to be ineffective in [15]. This is because both the detection probability, $\mathrm{P_D}$, and false alarm probability, $\mathrm{P_F}$, are determined by the same threshold. As a result, a higher value of $\mathrm{P_D}$ also implies a higher value of $\mathrm{P_F}$ and a smaller value of $\mathrm{P_D}$ also implies a smaller value of $\mathrm{P_F}$, and vice versa. So, to effectively assess the performance of AD, BS must be included for performance evaluation. The 3-D ROC analysis recently developed in [15] can be used for this purpose where AD can be evaluated by four categories of functionalities, such as TD, BS, joint TD and BS (TDBS), and ODP.

Several novelties which can also be considered as contributions are summarized in the following.

1) A new version of CEM using SBR as a constraint, SBR-CEM, is developed.
2) A new concept of DVT is introduced to extend SBR-CEM to UST-CEM and then to CEM-AD.
3) The OSP-AD in [13] can be rederived by DVT through extending OSP-TD to UST-OSP.
4) The LRaSMD model is introduced into CEM-AD to improve AD performance.
5) Two new criteria of SC, FSC and VSC, are designed for anomaly extraction.

The remainder of this article is organized as follows. Section II provides a survey of related works. Section III reviews CEM and reinterprets the minimum variance as SBR and also derives an SBR-based CEM which yields UST-CEM. Section IV develops DVT that can be used to derive CEM-AD. Section V proves two theorems that derive CEM-AD and OSP-AD via DVT. Section VI described the LRaSMD model produced by GoDec that can be used to develop CEM-AD and OSP-AD. Section VII develops two criteria of SC that can separate anomalies from noise in the GoDec generated sparse space. Section VIII provides 3-D ROC curve-derived detection measures to evaluate AD performance in terms of TD, BS, joint TD and BS, and overall detection performance. Section IX gives brief descriptions of three popularly used real hyperspectral images used for experiments and details the experimental results and discussions along with a detailed comparative analysis among five types of tested ADs. Finally, Section X concludes major contributions of this article.

II. Related Works

Chang [1] was first to make HTD and HAD dual theories. In particular, it showed how the GLRT can also be converted

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from HTD to the well-known AD, referred to as RX-AD developed by Reed and Xiaoli in [4] for HAD. In a traditional approach [5], [6] which used GLRT to derive RX-AD, it assumed that there was a secondary dataset which can be used to specify the target shape as a known signature so that the target amplitude can be estimated by maximum likelihood estimator to derive RX-AD. Similar to [5], [6], Liu *et al.* [16] also used a binary composite hypothesis testing approach to develop two equivalent GLRT-based adaptive ADs with the assumption that an anomalous pixel to be detected occupies multiple pixels with an unknown pattern that can be characterized by Gaussian distribution. Interestingly, the unknown new pattern used in [16] followed an idea similar to the dummy variable, $\mathbf{t}^{\mathrm{UST}}$, discussed in the introduction. Unfortunately, the anomaly detector developed in [16] was still confined to GLRT in which case, it can be considered a variant of GLRT. More specifically, there are two crucial differences. One is that the unknown pattern in [16] must be Gaussian distributed so that maximum likelihood estimation can be used to derive GLRT compared with $\mathbf{t}^{\mathrm{UST}}$ which is deterministic and can be any data sample. The other is that if an anomaly is present, it cannot occupy too many pixels; otherwise, it will become a known visible target by visual inspection. In this case, an anomaly does not have a sufficient number of pixels to form a pattern or to constitute a Gaussian distribution. Accordingly, assuming anomalies formed as patterns or making prior Gaussian distribution on anomalies in [16] seems inappropriate.

In HTD, the targets to be detected are assumed to be known. Therefore, its main task is how to improve TD. By contrast, in HAD, the targets to be detected are anomalies which are not supposed to be known. Under such a blind environment, HAD must address not only the issue of TD but also the issue of BS. This leads to a very important but crucial process required by AD, how to deal with BS. So, in recent years, the center piece of AD has been focused on the BKG issue. Several specific approaches have been investigated to model BKG for AD in the literature and briefly reviewed as follows.

One commonly used approach is LRaSR which decomposes the original data space $\mathbf{X}$ into $\mathbf{X} = \mathbf{DS} + \mathbf{E}$ using regularization parameters to constrain the sparse space $\mathbf{E}$ with $\mathbf{D}$ and $\mathbf{S}$ being considered as the BKG dictionary and representation coefficients [17], [18]. In other words, $\mathbf{DS}$ and $\mathbf{E}$ represent BKG and anomalies, respectively. Another is robust principal component analysis (RPCA) developed in [19] which can be used as a blind source separation technique to decompose the data matrix $\mathbf{X}$ into $\mathbf{X} = \mathbf{L} + \mathbf{S}$ in a matrix decomposition. Similarly, analogous to RPCA, a third approach is RSL developed in [20], [21] and [22], which treated an outlier as an additive sparse corruption via sparse+low-rank (S+LR) matrix decomposition. With this interpretation, RPCA can be considered as a variant of RSL via S+LR formulation.

However, the aforementioned models only offer separation of these two spaces with anomalies indeed sandwiched between BKG and noise. For example, RPCA and RSL via S+LR do not distinguish anomalies from noise or outliers in separate component spaces. Most recently, to address noise issue, a CDA was further developed to actually extend RPCA and RSL via S+LR to a three-component model which decomposes PCs, ICs, and noise component into three separate component spaces [23].

As an alternative to the above models, a more attractive model is LRaSMD produced by GoDec [14], which separates the noise in a third separate component space as a linear decomposition of a low rank matrix $\mathbf{X} = \mathbf{L} + \mathbf{S} + \mathbf{N}$. In order to better detect anomalies, GoDec did not use the commonly used parameters to regularize the sparse representation $\mathbf{S}$. Instead, it introduced the concept of SC to find an SC-constrained anomaly space from which anomalies can be extracted. This unique property was not found in any low rank and sparse representation-based approach. Interestingly, it is also this SC that makes LRaSMD very competitive. Unfortunately, GoDec did not discuss how SC works and how SC was determined. This article explores its great potential in enhancement of AD. In particular, two SC criteria, referred to as FSC and VSC, are specifically designed for the need of CEM-AD as well as OSP-AD. As a result, CEM-AD and OSP-AD using LRaSMD along with SC can significantly improve AD performance.

III. Constrained Energy Minimization

CEM was originally developed to detect targets at subpixel level. The idea arises from the linearly adaptive beamforming [24] by identifying signal arriving from one specific direction as desired signal source, $\mathbf{d}$, via designing a linear FIR filter to preserve the desired signal while minimizing the output energy of signal sources from other undesired directions.

A. CEM New Reinterpretation

Suppose that a hyperspectral image is represented by a collection of image pixel vectors, denoted by $\{\mathbf{r}_i\}_{i=1}^{N}$ with $\mathbf{r}_i = (r_{i1}, r_{i2}, \ldots, r_{iL})^T$ being an L-dimensional pixel vector and L being the total number of spectral bands. More specifically, let original HSI sample space be Ω_{data}, where $\Omega_{\mathrm{data}} = \{\mathbf{r}_n\}_{n=1}^{N}$ and $\mathbf{r}_n \in \Re^{L\times 1}$ be an nth data sample with L bands. Let $\delta(\mathbf{r}_n)$ be a linear filter specified by a weighting vector $\mathbf{w} = (w_1, w_2, \ldots, w_L)^T$ as

$$\delta(\mathbf{r}_n) = \mathbf{w}^T\mathbf{r}_n. \tag{1}$$

Then, its averaged output energy over the entire image scene is represented by

$$\frac{1}{N}\sum_{n=1}^{N}\delta^2(\mathbf{r}_n) = \mathbf{w}^T\mathbf{R}\mathbf{w} \tag{2}$$

where $\mathbf{R} = (1/N)\sum_{n=1}^{N}\mathbf{r}_n\mathbf{r}_n^T$ is the sample autocorrelation matrix, N is the total number of data samples, and L is the number of spectral bands. Now, the following constrained optimization problem can be formulated as follows:

$$\min_{\mathbf{w}}\left\{\mathbf{w}^T\mathbf{R}\mathbf{w}\right\} \quad \text{s.t. } \mathbf{d}^T\mathbf{w} = 1. \tag{3}$$

Using a Lagrange multiplier λ, we minimize the following constrained problem:

$$J(\mathbf{w}, \lambda) = \mathbf{w}^T\mathbf{R}\mathbf{w} + \lambda\left(\mathbf{d}^T\mathbf{w} - 1\right). \tag{4}$$

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Differentiating (4) with respect to $\mathbf{w}$ and λ, respectively, we have

$$\left.\frac{\partial J(\mathbf{w},\lambda)}{\partial \mathbf{w}}\right|_{\mathbf{w}^{\mathrm{CEM}}} = 2\mathbf{R}\mathbf{w}^{\mathrm{CEM}} + \lambda\mathbf{d} = 0 \Rightarrow \mathbf{w}^{\mathrm{CEM}} = -\frac{\lambda}{2}\mathbf{R}^{-1}\mathbf{d} \tag{5}$$

and

$$\left.\frac{\partial J(\mathbf{w},\lambda)}{\partial \lambda}\right|_{\lambda^{\mathrm{CEM}}} = \mathbf{d}^T\mathbf{w}^{\mathrm{CEM}} = 1. \tag{6}$$

Using (6), we reexpress (5) as

$$\mathbf{d}^T\mathbf{w}^{\mathrm{CEM}} = -\frac{\lambda^{\mathrm{CEM}}}{2}\mathbf{d}^T\mathbf{R}^{-1}\mathbf{d} = 1 \Rightarrow \lambda^{\mathrm{CEM}} = -\frac{2}{\mathbf{d}^T\mathbf{R}^{-1}\mathbf{d}}. \tag{7}$$

By substituting (7) into (5), we can derive

$$\mathbf{w}^{\mathrm{CEM}} = \mathbf{R}^{-1}\mathbf{d}\left(\mathbf{d}^T\mathbf{R}^{-1}\mathbf{d}\right)^{-1}. \tag{8}$$

Substituting (8) into (1) also yields the CEM detector, $\delta^{\mathrm{CEM}}(\mathbf{r})$ developed in [9] and given by

$$\delta_{\mathbf{d}}^{\mathrm{CEM}}(\mathbf{r}) = \left(\mathbf{w}^{\mathrm{CEM}}\right)^T\mathbf{r} = \frac{\mathbf{d}^T\mathbf{R}^{-1}\mathbf{r}}{\mathbf{d}^T\mathbf{R}^{-1}\mathbf{d}} \tag{9}$$

where the subscript "$\mathbf{d}$" is included to emphasize the use of the desired target signature $\mathbf{d}$. The minimum variance resulting from (9) can be obtained as

$$\left(\mathbf{w}^{\mathrm{CEM}}\right)^T\mathbf{R}\mathbf{w}^{\mathrm{CEM}} = \left(\mathbf{d}^T\mathbf{R}^{-1}\mathbf{d}\right)^{-1}. \tag{10}$$

Now, if we let Δ be all data samples specified by the target signature $\mathbf{d}$, then we can define a new correlation matrix $\mathbf{R}_{\Omega_{\mathrm{data}}-\Delta}$ by removing all data samples in Δ from the entire data sample space as

$$\mathbf{R}_{\Omega_{\mathrm{data}}-\Delta} = \frac{1}{|\Omega_{\mathrm{data}} - \Delta|}\sum_{\Omega_{\mathrm{data}}-\Delta}\mathbf{r}_n\mathbf{r}_n^T \in \mathbb{R}^{L\times L} \tag{11}$$

where $|\Omega_{\mathrm{data}} - \Delta|$ is the size of $\Omega_{\mathrm{data}} - \Delta$. In this case, $\mathbf{R}_{\Omega_{\mathrm{data}}-\Delta}$ is formed by all data samples not specified by $\mathbf{d}$ which can be considered as BKG samples so that

$$\mathrm{SBR}\left(\delta_{\mathbf{d}}^{\mathrm{CEM}}\right) = \mathbf{d}^T\mathbf{R}_{\Omega_{\mathrm{data}}-\Delta}^{-1}\mathbf{d} \tag{12}$$

can be interpreted as signal ($\mathbf{d}$)-to-BKG (Δ) ratio (SBR) of $\delta_{\mathbf{d}}^{\mathrm{CEM}}$. Replacing $\mathbf{R}^{-1}$ in (10) with $\mathbf{R}_{\Omega_{\mathrm{data}}-\Delta}^{-1}$ gives rise to

$$\left(\mathbf{w}^{\mathrm{CEM}}\right)^T\mathbf{R}_{\Omega_{\mathrm{data}}-\Delta}^{-1}\mathbf{w}^{\mathrm{CEM}} = \left(\mathbf{d}^T\mathbf{R}_{\Omega_{\mathrm{data}}-\Delta}^{-1}\mathbf{d}\right)^{-1} \tag{13}$$

which implies that minimum variance of (10) is indeed the same as SBR in (13). This is also confirmed by [2, eqs. (50)] which is SNR derived by the *R*-SNR approach and also defined in [25, p. 57].

Since inverting the correlation matrix $\mathbf{R}$ is simply equivalent to suppressing the interfering effects of data samples that forms $\mathbf{R}$ on detection of $\mathbf{d}$. So, technically speaking, CEM should use (11) instead of (3) to have better target detection performance as shown in [3] and [26]. However, due to the fact that the size of Δ is generally very small, the difference between (10) and (13) is not much appreciable. This is why CEM uses $\mathbf{R}^{-1}$ rather than $\mathbf{R}_{\Omega_{\mathrm{data}}-\Delta}^{-1}$. In this case, (12) is

$$\mathrm{SBR}\left(\delta_{\mathbf{d}}^{\mathrm{CEM}}\right) = \mathbf{d}^T\mathbf{R}_{\Omega_{\mathrm{data}}}^{-1}\mathbf{d} \approx \mathbf{d}^T\mathbf{R}^{-1}\mathbf{d}^T \tag{14}$$

which further shows that minimum variance in (10) is actually SBR in (14).

So, in this section, the novelty is to interpret the minimum variance resulting from CEM as SBR derived in (14).

B. SBR-CEM

Since AD is not supposed to know targets it detects, there is no specific $\mathbf{d}$ available to be used for detection. So, a direct use of CEM in AD is not applicable. This section derives an SBR-CEM which paves the way for the DVT in Section III to be used to derive CEM-AD.

Interestingly, from (6), we can use $\mathbf{t}^{\mathrm{UST}}$ to replace $\mathbf{d}$ in (3) as the constraint $(\mathbf{t}^{\mathrm{UST}})^T\mathbf{w} = 1$ to derive λ in (7) which turns out to be the constant $((\mathbf{t}^{\mathrm{UST}})^T\mathbf{R}^{-1}\mathbf{t}^{\mathrm{UST}})^{-1}$ in CEM to account for the accurate target signature abundance fraction of $\mathbf{t}^{\mathrm{UST}}$ to be detected in the data samples.

$$\delta_{\mathbf{t}^{\mathrm{UST}}}^{\mathrm{UST\text{-}CEM}}(\mathbf{r}) = \frac{\left(\mathbf{t}^{\mathrm{UST}}\right)^T\mathbf{R}^{-1}\mathbf{r}}{\left(\mathbf{t}^{\mathrm{UST}}\right)^T\mathbf{R}^{-1}\mathbf{t}^{\mathrm{UST}}}. \tag{15}$$

Since we do not know $\mathbf{t}^{\mathrm{UST}}$ and $((\mathbf{t}^{\mathrm{UST}})^T\mathbf{R}^{-1}\mathbf{t}^{\mathrm{UST}})^{-1}$ is a constant, it does not make sense to include this unknown constant in the detector design. On way to do so is that we can simply remove this constant from (15) to yield a UST-CEM defined as

$$\delta_{\mathbf{t}^{\mathrm{UST}}}^{\mathrm{UST\text{-}CEM}}(\mathbf{r}) = \left(\mathbf{t}^{\mathrm{UST}}\right)^T\mathbf{R}^{-1}\mathbf{r}. \tag{16}$$

However, it is easily said than done. This is because (16) is different from (9) in the sense that $\mathbf{t}^{\mathrm{UST}}$ in (16) is assumed to be an unknown specific target signature without prior knowledge compared to $\mathbf{d}$ used in (9) which is known provided *a priori*. The issue is that is (16) a valid form? If it is, how can we derive (16) mathematically?

According to (14), we can use (14) as the SBR constraint for CEM, i.e., $(\mathbf{t}^{\mathrm{UST}})^T\mathbf{w} = \mathrm{SBR} = (\mathbf{t}^{\mathrm{UST}})^T\mathbf{R}^{-1}\mathbf{t}^{\mathrm{UST}}$ to replace $\mathbf{d}^T\mathbf{w} = 1$ used in (3). The resulting constrained optimization problem is then given by

$$\begin{aligned}&\min_{\mathbf{w}}\ \left\{\mathbf{w}^T\mathbf{R}\mathbf{w}\right\}\\ &\text{s.t. } \left(\mathbf{t}^{\mathrm{UST}}\right)^T\mathbf{w} = \mathrm{SBR} = \left(\mathbf{t}^{\mathrm{UST}}\right)^T\mathbf{R}^{-1}\mathbf{t}^{\mathrm{UST}}.\end{aligned} \tag{17}$$

The solution to (17) is called SBR-CEM which can be proved in the following theorem.

Theorem 1:

$$\delta_{\mathbf{t}^{\mathrm{UST}}}^{\mathrm{SBR\text{-}CEM}}(\mathbf{r}) = \left(\mathbf{w}^{\mathrm{SBR\text{-}CEM}}\right)^T\mathbf{r} = \left(\mathbf{t}^{\mathrm{UST}}\right)^T\mathbf{R}^{-1}\mathbf{r}. \tag{18}$$

Proof: By introducing a Lagrange multiplier λ into (17), (4) becomes

$$J(\mathbf{w},\lambda) = \mathbf{w}^T\mathbf{R}\mathbf{w} + \lambda\left(\left(\mathbf{t}^{\mathrm{UST}}\right)^T\mathbf{w} - \left(\mathbf{t}^{\mathrm{UST}}\right)^T\mathbf{R}^{-1}\mathbf{t}^{\mathrm{UST}}\right). \tag{19}$$

Differentiating (19) with respect to $\mathbf{w}$ and λ, respectively, we have

$$\left.\frac{\partial J(\mathbf{w},\lambda)}{\partial \mathbf{w}}\right|_{\mathbf{w}^{\mathrm{SBR\text{-}CEM}}} = 0 \Rightarrow \mathbf{w}^{\mathrm{SBR\text{-}CEM}} = -\frac{\lambda}{2}\mathbf{R}^{-1}\mathbf{t}^{\mathrm{UST}} \tag{20}$$

$$\left.\frac{\partial J(\mathbf{w},\lambda)}{\partial \lambda}\right|_{\lambda^{\mathrm{SBR\text{-}CEM}}} = 0$$

$$\Rightarrow \left(\mathbf{t}^{\mathrm{UST}}\right)^T\mathbf{w}^{\mathrm{SBR\text{-}CEM}} = \mathrm{SBR} = \left(\mathbf{t}^{\mathrm{UST}}\right)^T\mathbf{R}^{-1}\mathbf{t}^{\mathrm{UST}}. \tag{21}$$

Using (19), we reexpress (18) as

$$\begin{aligned}\left(\mathbf{t}^{\mathrm{UST}}\right)^T\mathbf{w}^{\mathrm{SBR\text{-}CEM}} &= -\frac{\lambda^{\mathrm{CEM}}}{2}\left(\mathbf{t}^{\mathrm{UST}}\right)^T\mathbf{R}^{-1}\mathbf{t}^{\mathrm{UST}}\\ &= \left(\mathbf{t}^{\mathrm{UST}}\right)^T\mathbf{R}^{-1}\mathbf{t}^{\mathrm{UST}} \Rightarrow \lambda^{\mathrm{SBR\text{-}CEM}} = -2.\end{aligned} \tag{22}$$

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Substituting (21) into (19) yields

$$\mathbf{w}^{\text{SBR-CEM}} = \mathbf{R}^{-1}\mathbf{t}^{\text{UST}}. \tag{23}$$

By substituting (23) into (1), the original CEM in (9) becomes SBR-constrained CEM (SBR-CEM), $\delta^{\text{SBR-CEM}}_{\mathbf{t}^{\text{UST}}}(\mathbf{r})$, given by (18) which is exactly the same (16). ■

In light of (16) and (18), we can assume that $\mathbf{t}^{\text{UST}}$ in (16) and (18) is actually a UST signature and further define a CEM-AD, $\delta^{\text{CEM-AD}}(\mathbf{r})$, by maximizing the following SBR-CEM over all possible UST

$$\delta^{\text{CEM-AD}}(\mathbf{r}) = \max_{\mathbf{t}^{\text{UST}} \in \Omega_{\text{data}}} \delta^{\text{SBR-CEM}}_{\mathbf{t}^{\text{UST}}}(\mathbf{r}) \tag{24}$$

where $\mathbf{t}^{\text{UST}}$ is introduced as a dummy variable to be discussed in Section III and is vanished in the same way that RX-AD was derived as max SNR [1, (26) in Ths. 3]. It is also worth noting that if we replace the UST signature, $\mathbf{t}^{\text{UST}}$, in (18) with the known target signature $\mathbf{d}$, (18) becomes (9) without normalization constant $\mathbf{d}^T\mathbf{R}^{-1}\mathbf{d}$, i.e., $\mathbf{d}^T\mathbf{R}^{-1}\mathbf{r}$ where the normalization constant $\mathbf{d}^T\mathbf{R}^{-1}\mathbf{d}$ turns out the SBR in (14).

The main novelty of this section is the development of a new version of CEM constrained by SBR, SBR-CEM.

IV. Dummy Variable Trick

Before stating the DVT, a simple toy example may help to enlighten the notion of DVT. Suppose that there is a farmer who owns a total of 17 cows to be given to his three sons. The eldest one will receive 1/2 of his 17 cows followed by the second and youngest sons who will receive 1/3 and 1/9 of his 17 cows, respectively. Obviously, 17 is a prime and cannot be divisible. To solve this problem, the smart farmer introduces a hypothetic cow, denoted by ξ, as a dummy cow into 17 cows to make it 18 cows. As a result of 18 cows, the eldest, second, and youngest sons will receive exactly their fair share proportions, 9, 6, and 2 cows, respectively, added up to 17 cows. The remaining one cow happens to be the dummy cow, ξ, which is the nonexistent hypothetic cow to account for 1/18 of 18 cows that will be given to the farmer himself who ends up with not owning any cow.

Now, we can use (18) to present DVT which can be stated as follows.

Dummy Variable Trick: For a problem A which cannot be solved directly, the DVT introduces an auxiliary variable ξ to reformulate a new problem B which is solvable where ξ is used as a liaison to bridge problems A and B and will be vanished after the problem is solved. Because of its functionality, ξ is considered as a dummy variable. ■

It is worth noting that the dummy variable used by DVT can be also regarded as a latent variable, hidden variable or unobserved variable widely used in pattern recognition [13]. Interestingly, the concept of DVT has been used in several applications but its potential usefulness has never been realized.

1) Extension of binary Huffam coding to p-ary Huffman coding: the most well-known example is to extend a binary Huffman coding to p-ary Huffman coding with $p > 2$ where dummy variables are introduced to account for optimal coding [27, pp. 336–340]. Since each internal note of a p-ray coding has more than 2 subtrees, the p-ary tree constructed by a p-ary Huffman code may not be complete in the sense that there may not have sufficient codewords to fill in all nodes and leaves of the tree. In this case, unencoded nodes must be leaves and should not be internal nodes. In order to address this issue, we need to introduce dummy source alphabets so that these dummy alphabets will be treated as source alphabets with zero probability in such a way that they will have longest source codewords which do not exist. The key issue is how many dummy variables are required to do so when we construct a p-ary Huffman code for any positive integer p. Let q be the number of the first set of source alphabets to be combined to reduce original probabilities $\{p_1, p_2, \ldots, p_n\}$ to $\{p_1^{(1)}, p_2^{(1)}, \ldots, p_{n^{(1)}}^{(1)}\}$. According to [27, eqs. (11.2) and (11.3), p. 338], q must satisfy the constraints $q \in \{2, 3, \ldots, p\}$ and $n \equiv q \bmod (p-1)$, i.e., $n = m(p-1) + q$ for some nonnegative integer m. Then, the number of dummy variables is $\{x_1, x_2, \ldots, x_{p-q}\}$. In this case, $n_1^{(1)} = m(p-1) + 1$. As an example, for $n = 6$ and $p = 3$, n can be expressed by (6) as $n = 2 \cdot 2 + 2$ with $m = 2$ and $q = 2$. This implies that the number of the dummy variable is $p - q = 3 - 2 = 1$.
2) Channel capacity computation: a second most important application is channel capacity computation problem [28, Ths. 1a]. According to its definition, the mutual information is given by

$$I(X; Y) = \sum_{j=1}^{J} p_j \sum_{k=1}^{K} Q_{k|j} \log \frac{Q_{k|j}}{q_k} \tag{25}$$

with the channel input space, the channel output space, and the forward channel specified by X with the probability vector $\mathbf{p} = (p_1, p_2, \ldots, p_J)^T$ and Y with the probability vector $\mathbf{q} = (q_1, q_2, \ldots, q_k)^T$ and $\{Q_{k|j}\}_{j=1,k=1}^{J,K}$, respectively. Let

$$J\left(\mathbf{p}, \{Q_{k|j}\}, \hat{\mathbf{P}}\right) = \sum_{j=1}^{J} p_j \sum_{k=1}^{K} Q_{k|j} \log \frac{\hat{P}_{j|k}}{q_k} \tag{26}$$

where $\hat{\mathbf{P}} = \{\hat{P}_{j|k}\}_{k=1,j=1}^{K,J}$ is a backward channel transition matrix. Then, the channel capacity defined as $C = \max_{\mathbf{p}} I(X, Y)$ can be now reformulated as the following double maxima problem:

$$C = \max_{\mathbf{p}} I(X, Y) = \max_{\mathbf{p}} \max_{\hat{\mathbf{P}}} \left\{ J\left(\mathbf{p}, \hat{\mathbf{P}}\right) \right\} \tag{27}$$

where $\hat{\mathbf{P}}$ is used as a dummy variable ξ considered in DVT. The optimal solution $\hat{\mathbf{P}}^*$ in (27) is given by $\hat{P}^*_{j|k} = (p_j Q_{k|j})/q_k^*$ with $q_k^* = \sum_{j=1}^{J} p_j Q_{k|j}$ which is eventually vanished because $\hat{\mathbf{P}}^*$ is actually determined by the given forward channel $\{Q_{k|j}\}_{j=1,k=1}^{J,K}$ and channel input probability vector $\mathbf{p} = (p_1, p_2, \ldots, p_J)^T$.
3) Rate distortion function: a third most important application is rate distortion function computation [28, Ths. 4a]. Given its definition, for any distortion D, the rate

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distortion function $R(D)$ is expressed as

$$R(D) = \min_{\{Q_{k|j}\}_{j=1,k=1}^{J,K}} \{I(X;Y)\} \quad \text{s.t.} \sum_{j=1}^{J}\sum_{k=1}^{K} p_j Q_{k|j}\rho_{jk} \le D \tag{28}$$

where ρ_{jk} is distortion between the jth channel input and kth channel output and $D = \sum_{j=1,k=1}^{J,K} p_j Q^*_{k|j}\rho_{jk}$ with $Q^*_{k|j}$ achieving the minimum in (28). Similar to channel capacity computation problem, a Lagrange multiplier penalty function can be defined as

$$F\left(\mathbf{p}, \left\{Q_{k|j}\right\}, \mathbf{q}\right) = \sum_{j=1}^{J} p_j \sum_{k=1}^{K} Q_{k|j} \log \frac{Q_{j|k}}{q_k} - s \sum_{j=1,k=1}^{J,K} p_j Q_{k|j}\rho_{jk}. \tag{29}$$

By virtual of (29), we reformulate (28) as a double minima problem given by

$$R(D) = sD + \min_{\mathbf{q}} \min_{\{Q_{k|j}\}} F\left(\mathbf{p}, \left\{Q_{k|j}\right\}, \mathbf{q}\right) \tag{30}$$

with s used as the Lagrange multiplier to constrain $\sum_{j=1}^{J}\sum_{k=1}^{K} p_j Q_{k|j}\rho_{jk} = D$ where the probability output vector $\mathbf{q} = (q_1, q_2, \ldots, q_K)^T$ is used as a dummy variable ξ considered in DVT and will be determined and vanished by the optimal solution $q_k^* = \sum_{j=1}^{J} p_j Q_{k|j}$.

4) Compressive sensing: a fourth interesting application is compressive sensing where there are two matrices, sensing matrix $\mathbf{\Phi}$ and sparse matrix $\mathbf{\Psi}$ used to acquire signals as

$$\bar{\mathbf{r}} = \mathbf{\Phi}\mathbf{\Psi}\mathbf{r} + \mathbf{n} \tag{31}$$

where $\mathbf{r} \in \Re^{L\times 1}$ and $\bar{\mathbf{r}} \in \Re^{m\times 1}$ are the original hyperspectral signal vector and its compressed measurement vector, respectively, and $\mathbf{\Phi} \in \Re^{m\times L}$ is the compressed sampling matrix, $\mathbf{\Psi} \in \Re^{L\times L}$ is the sparse representation basis, and $\mathbf{n} \in \Re^{m\times 1}$ is a noise term. According to the "*universality*" property of random sensing described in [29], the sparse basis, $\mathbf{\Psi}$, is only needed for signal recovery. Consequently, if we are not interested in signal recovery such as hyperspectral image classification [30], (31) can be further reduced to a simplified model as

$$\bar{\mathbf{r}} = \mathbf{\Phi}_m\mathbf{r} + \mathbf{n} \tag{32}$$

where $\mathbf{\Phi}_m$ represents a random sampling matrix using m compressively sensed bands. Equation (32) has an extremely important implication that for algorithms performed in the compressively sensed domain, an optimal performance can be actually achieved with no need of identifying the sparse matrix $\mathbf{\Psi}$ which can be considered as dummy variable ξ used in DVT.

5) Extension of $\delta_{\mathbf{t}}^{\text{GLRT}}(\mathbf{r}) = ((\mathbf{t}^T\mathbf{K}^{-1}(\mathbf{r}-\boldsymbol{\mu}))^2)/(\mathbf{t}^T\mathbf{K}^{-1}\mathbf{t})$ in [1, eqs. (7)] to AD $\delta^{\text{RX-AD}}(\mathbf{r}) = (\mathbf{r}-\boldsymbol{\mu})^T\mathbf{K}^{-1}(\mathbf{r}-\boldsymbol{\mu})$ in [4]: a fifth application is the one closest to extension of CEM to CEM-AD via $\delta^{\text{CEM-AD}}(\mathbf{r}) = \max_{\mathbf{t}^{\text{UST}}} \delta_{\mathbf{t}^{\text{UST}}}^{\text{SBR-CEM}}(\mathbf{r})$ in (24) to be proved in Theorem 2 in Section V. In other words, GLRT can be also extended to UST-GLRT, $\delta_{\mathbf{t}^{\text{UST}}}^{\text{GLRT}}(\mathbf{r})$ [1, eqs. (17)] and then to $\delta^{\text{RX-AD}}(\mathbf{r}) = \max_{\mathbf{t}^{\text{UST}}} \delta_{\mathbf{t}^{\text{UST}}}^{\text{GLRT}}(\mathbf{r})$ [1, eqs. (26) in Ths. 3]. More specifically, their corresponding connection can be described by the following relationships:

$$\begin{aligned}\delta_{\mathbf{t}}^{\text{GLRT}}(\mathbf{r}) &\Rightarrow \delta_{\mathbf{t}^{\text{UST}}}^{\text{GLRT}}(\mathbf{r}) \Rightarrow \delta^{\text{RX-AD}}(\mathbf{r}) \text{ in [4]}\\ \delta_{\mathbf{d}}^{\text{CEM}}(\mathbf{r}) &\Rightarrow \delta_{\mathbf{t}^{\text{UST}}}^{\text{UST-CEM}}(\mathbf{r}) \Rightarrow \delta^{\text{CEM-AD}}(\mathbf{r}) \text{ in (24)}\\ &= \delta^{\text{R-AD}}(\mathbf{r}) \text{ in [31].}\end{aligned}$$

It should be noted that R-AD was taken for granted in [8], [31]. Its mathematical proof is not trivial but rather challenging. Theorem 2 in Section V proves that R-AD is indeed a special case of CEM-AD via DVT.

Main novelty of this section is the development of DVT via the SBR-CEM derived in Section II-B and further provides a sixth application of DVT in AD.

V. CEM and OSP ADs

The DVT developed in Section IV paved the way for CEM and OSP-TD be extended to their respective CEM-AD and OSP-AD.

A. CEM-AD

Recall that CEM requires the prior target knowledge of $\mathbf{d}$. To extend CEM to an AD, this $\mathbf{d}$ can be taken care of by $\mathbf{t}^{\text{UST}}$ via DVT as follows.

Theorem 2:

$$\delta_{\kappa}^{\text{CEM-AD}}(\mathbf{r}) = \kappa\mathbf{r}^T\mathbf{R}^{-1}\mathbf{r}, \quad \text{for some constant } \kappa. \tag{33}$$

Proof: Using (18) by DVT, we derive the following inequality by Schwarz's inequality

$$\begin{aligned}\left(\mathbf{t}^{\text{UST}}\right)^T\mathbf{R}^{-1}\mathbf{r} &= \left\|\left(\mathbf{t}^{\text{UST}}\right)^T\mathbf{R}^{-1/2}\mathbf{R}^{-1/2}\mathbf{r}\right\|\\ &\le \left\|\left(\mathbf{t}^{\text{UST}}\right)^T\mathbf{R}^{-1/2}\right\|\left|\mathbf{R}^{-1/2}\mathbf{r}\right| = \left\|\mathbf{R}^{-1/2}\mathbf{r}\right\|^2\end{aligned} \tag{34}$$

where the equality holds if and only if $\mathbf{t}^{\text{UST}} = \kappa\mathbf{r}$ for some constant κ. By virtue of (34), we obtain

$$\max_{\mathbf{t}^{\text{UST}}\in\Omega_{\text{data}}} \left(\mathbf{t}^{\text{UST}}\right)^T\mathbf{R}^{-1}\mathbf{r} = \kappa\mathbf{r}^T\mathbf{R}^{-1}\mathbf{r} \tag{35}$$

which turns out to be the R-SNR detector derived by SNR in [1, eqs. (48)]. Setting $\kappa = 1$ yields

$$\delta_{\kappa=1}^{\text{CEM-AD}}(\mathbf{r}) = \mathbf{r}^T\mathbf{R}^{-1}\mathbf{r} \tag{36}$$

which is exactly R-AD derived in [31].

In contrast, by setting $\kappa = ((\mathbf{t}^{\text{UST}})^T\mathbf{R}^{-1}\mathbf{t}^{\text{UST}})^{-1}$, we can derive

$$\delta_{\kappa=\left(\left(\mathbf{t}^{\text{UST}}\right)^T\mathbf{R}^{-1}\mathbf{t}^{\text{UST}}\right)^{-1}}^{\text{CEM-AD}}(\mathbf{r}) = \frac{\mathbf{r}^T\mathbf{R}^{-1}\mathbf{r}}{\left(\mathbf{t}^{\text{UST}}\right)^T\mathbf{R}^{-1}\mathbf{t}^{\text{UST}}} \tag{37}$$

which becomes UST-CEM in (15).

It should be noted that if the original CEM in (9) is used for AD by replacing $\mathbf{d}$ with $\mathbf{r}$, it is always constant 1 regardless of data samples due to the constraint $\mathbf{d}^T\mathbf{w} = 1$ imposed on CEM in (9). This explains why CEM cannot be used for AD.

The CEM-AD in Theorem 2 cannot be derived directly from CEM without using Theorem 1. This is because DVT uses UST-CEM in (16) and SBR-CEM in (18) as a liaison to bridge

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CEM and CEM-AD, $\delta_{\kappa}^{\text{R-AD}}(\mathbf{r})$. However, it is interesting to note that the CEM-AD (33) derived in Theorem 2 from using SBR-constrained CEM, $\delta_{\mathbf{t}^{\text{UST}}}^{\text{SBR-CEM}}(\mathbf{r})$, via DVT turns out to be exactly the same as $\delta^{\text{R-AD}}(\mathbf{r})$ derived from SNR using data sphering without using DVT [1, eqs. (48)]. This shows that two different approaches can arrive at the same R-AD, $\delta^{\text{R-AD}}(\mathbf{r})$.

B. OSP-AD

Interestingly, DVT can also be used to prove OSP-AD derived in [11].

Let L be the total number of spectral bands and $\mathbf{r}$ be an L-dimensional data sample vector. Assume that $\mathbf{m}_1, \mathbf{m}_2, \ldots, \mathbf{m}_p$ are p spectral signatures of interest which are linearly mixed by their corresponding abundance fractions given by $\alpha_1, \alpha_2, \ldots, \alpha_p$. Let $\mathbf{m}_p = \mathbf{d}$ be the single desired signature with its proportional abundance fraction α_p without loss of generality and the remaining signatures, $\mathbf{m}_1, \mathbf{m}_2, \ldots, \mathbf{m}_{p-1}$, as undesired spectral signatures to form an undesired spectral signature matrix denoted by $\mathbf{U}_{p-1} = [\mathbf{m}_1 \mathbf{m}_2, \ldots, \mathbf{m}_{p-1}]$ with $\boldsymbol{\gamma} = (\alpha_1, \alpha_2, \ldots, \alpha_{p-1})^T$ being its corresponding abundance fraction vector. Then, an OSP detector, $\delta_{\mathbf{d}}^{\text{OSP}}(\mathbf{r})$, was derived in [9]

$$\delta_{\mathbf{d}}^{\text{OSP}}(\mathbf{r}) = \mathbf{d}^T P_{\mathbf{U}_{p-1}}^{\perp} \mathbf{r} \tag{38}$$

where $P_{\mathbf{U}_{p-1}}^{\perp}$ is given by

$$P_{\mathbf{U}_{p-1}}^{\perp} = \mathbf{I} - \mathbf{U}_{p-1} \mathbf{U}_{p-1}^{\#}. \tag{39}$$

With $\mathbf{U}_{p-1}^{\#} = (\mathbf{U}_{p-1}^T \mathbf{U}_{p-1})^{-1} \mathbf{U}_{p-1}^T$ being the pseudo-inverse of $\mathbf{U}_{p-1}$.

Now, we can also consider (38) as a UST-OSP defined as

$$\delta_{\mathbf{t}^{\text{UST}}}^{\text{UST-OSP}}(\mathbf{r}) = \left(\mathbf{t}^{\text{UST}}\right)^T P_{\mathbf{U}_{p-1}}^{\perp} \mathbf{r} \tag{40}$$

in the same manner by DVT that UST-CEM in (16) and SBR-CEM in (18) consider $\mathbf{t}^{\text{UST}}$ in (40) to be a UST signature without prior knowledge which will be vanished after all. In particular, following a similar treatment used to prove Theorem 2, we can also derive the following theorem.

Theorem 3:

$$\delta_{\kappa}^{\text{OSP-AD}}(\mathbf{r}) = \kappa \mathbf{r}^T P_{\mathbf{U}_{p-1}}^{\perp} \mathbf{r}, \quad \text{for some constant } \kappa. \tag{41}$$

Proof: Using OSP in (40) via DVT, we can derive the following inequality by Schwarz's inequality

$$\begin{aligned} \left(\mathbf{t}^{\text{UST}}\right)^T P_{\mathbf{U}_{p-1}}^{\perp} \mathbf{r} &= \left\| \left(\mathbf{t}^{\text{UST}}\right)^T P_{\mathbf{U}_{p-1}}^{\perp} P_{\mathbf{U}_{p-1}}^{\perp} \mathbf{r} \right\| \\ &\le \left\| \left(\mathbf{t}^{\text{UST}}\right)^T P_{\mathbf{U}_{p-1}}^{\perp} \right\| \left\| P_{\mathbf{U}_{p-1}}^{\perp} \mathbf{r} \right\| \end{aligned} \tag{42}$$

where the equality holds if and only if

$$P_{\mathbf{U}_{p-1}}^{\perp} \mathbf{t}^{\text{UST}} = \kappa P_{\mathbf{U}_{p-1}}^{\perp} \mathbf{r} \Rightarrow \mathbf{t}^{\text{UST}} = \kappa \mathbf{r} \tag{43}$$

for some constant κ. Substituting (43) into (40), we obtain

$$\max_{\mathbf{t}^{\text{UST}} \in \Omega_{\text{data}}} \left(\mathbf{t}^{\text{UST}}\right)^T P_{\mathbf{U}_{p-1}}^{\perp} \mathbf{r} = \kappa \mathbf{r}^T P_{\mathbf{U}_{p-1}}^{\perp} \mathbf{r} \tag{44}$$

where $\mathbf{t}^{\text{UST}}$ is introduced as a dummy parameter variable and is vanished. Equation (44) turns out to be the OSP-AD derived in [11]. Setting $\kappa = 1$ and $\mathbf{U}_{p-1} = \mathbf{U}$ in (44) yields

$$\delta_{\kappa=1}^{\text{OSP-AD}}(\mathbf{r}) = \mathbf{r}^T P_{\mathbf{U}_{p-1}}^{\perp} \mathbf{r} \tag{45}$$

$$\begin{array}{ccccc} \text{CEM} & \rightarrow & \begin{array}{c}\text{UST-CEM}\\ \text{(DVT)}\end{array} & \rightarrow & \text{CEM-AD} \\ \dfrac{\mathbf{d}^T \mathbf{R}^{-1} \mathbf{r}}{\mathbf{d}^T \mathbf{R}^{-1} \mathbf{d}} & & \left(\mathbf{t}^{\text{UST}}\right)^T \mathbf{R}^{-1} \mathbf{r} & & \begin{array}{c}\kappa \mathbf{r}^T \mathbf{R}^{-1} \mathbf{r}\\ \text{Theorem 2}\end{array} \\ \updownarrow & & \mathbf{t}^{\text{UST}} = \text{dummy variable} & & \updownarrow \\ \mathbf{d}^T P_{\mathbf{U}}^{\perp} \mathbf{r} & & \left(\mathbf{t}^{\text{UST}}\right)^T P_{\mathbf{U}}^{\perp} \mathbf{r} & & \begin{array}{c}\kappa \mathbf{r}^T P_{\mathbf{U}}^{\perp} \mathbf{r}\\ \text{Theorem 3}\end{array} \\ \text{OSP-TD} & \rightarrow & \text{UST-OSP} & \rightarrow & \text{OSP-AD} \end{array}$$

Fig. 1. Relationship among CEM, OSP-TD, CEM-AD, and OSP-AD via DVT where USTCEM and USTOSP in the middle column are dummy variables.

which is exactly the same as the R-AD derived in [11, eqs. (18)].

Finally, Fig. 1 depicts the relationship among CEM, OSP-TD, CEM-AD, and OSP-AD via DVT where USTCEM and USTOSP for a given data sample vector **r, d**, and $\mathbf{t}^{\text{UST}}$ are the known desired target signature and UST signature as a dummy variable, respectively.

Two novelties derived from this section are to develop CEM-AD by DVT and rederive OSP-AD in [11] using DVT.

VI. LRaSMD by GoDec

In order for CEM-AD and OSP-AD to perform AD, they both use the concept of the UST $\mathbf{t}^{\text{UST}}$ as a liaison, $\delta_{\mathbf{t}^{\text{UST}}}^{\text{UST-CEM}}(\mathbf{r})/\delta_{\mathbf{t}^{\text{UST}}}^{\text{SBR-CEM}}(\mathbf{r})$ and $\delta_{\mathbf{t}^{\text{UST}}}^{\text{UST-OSP}}(\mathbf{r})$ to bridge CEM and CEM-AD, and OSP and OSP-AD. The major difference between $\delta^{\text{CEMAD}}(\mathbf{r})$ in (36) and $\delta^{\text{OSPAD}}(\mathbf{r})$ in (45) is that CEM-AD in (36) performs AD by inverting the sample correlation matrix $\mathbf{R}$ to suppress BKG, while OSP-AD in (45) performs AD by $P_{\mathbf{U}_{p-1}}^{\perp}$ to remove interfering effect resulting from the signatures in $\mathbf{U}_{p-1}$. So, if $\mathbf{U}_{p-1}$ is made up of BKG signatures, then OSP-AD indeed removes BKG as opposed to CEM suppressing BKG. In other words, there are two ways to resolve the BKG issue, that is, BS and BKG removal. Interestingly, CEM-AD and OSP-AD are designed particularly for these two purposes, respectively, i.e., CEM-AD for BS by inverting $\mathbf{R}$ and OSP-AD for BKG removal by $P_{\mathbf{U}_{p-1}}^{\perp}$ with $\mathbf{U}_{p-1}$ specified by BKG. The key issue is how to cope with BKG issue which has been a major stream in the recent research in AD.

This section considers the LRaSMD model produced by GoDec as

$$\mathbf{X} = \mathbf{L} + \mathbf{S} + \mathbf{N} \quad \text{s.t. } \text{rank}(\mathbf{L}) \le r, \quad \text{card}(\mathbf{S}) = \text{SC} \le k \tag{46}$$

which has been shown to perform better than the two-component decomposition, $\mathbf{X} = \mathbf{L} + \mathbf{S}$, produced by RPCA [19].

The idea of GoDec [14] is to use BRP to solve the following low-rank approximation:

$$\mathbf{L} = \mathbf{Q}_1 \left[\mathbf{R}_1 \left(\mathbf{A}_2^T \mathbf{Y}_1\right)^{-1} \mathbf{R}_2^T\right]^{1/(2q+1)} \mathbf{Q}_2^T \tag{47}$$

where $\mathbf{Q}_1, \mathbf{Q}_2, \mathbf{R}_1, \mathbf{R}_2$ are QR decomposition of $\mathbf{Y}_1, \mathbf{Y}_2$, and q is an empirically predetermined parameter. In addition,

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the model in (46) can be further solved by minimizing the following decomposition error:

$$\min_{\mathbf{L}_{N\times L},\mathbf{S}_{N\times L}} \|\mathbf{X}_{N\times L}-\mathbf{L}_{N\times L}-\mathbf{S}_{N\times L}\|_F^2 \tag{48}$$

$$\text{s.t. } \operatorname{rank}(\mathbf{L}_{N\times L})\le m \quad \text{and } \operatorname{card}(\mathbf{S}_{N\times L})\le jN \tag{49}$$

where $\|\cdot\|_F$ is the Frobenius matrix norm. Since (46) does not have an analytical solution, it must rely on a numerical algorithm, which is GoDec to find solutions by the following iterative processes:

$$\mathbf{L}_{N\times L}^{(t)} = \arg\min_{\operatorname{rank}(\mathbf{L}_{N\times L})\le m}\left\|\mathbf{X}_{N\times L}-\mathbf{L}_{N\times L}-\mathbf{S}_{N\times L}^{(t-1)}\right\|_F^2 \tag{50}$$

$$\mathbf{S}_{N\times L}^{(t)} = \arg\min_{\operatorname{card}(\mathbf{S}_{N\times L})\le jN}\left\|\mathbf{X}_{N\times L}-\mathbf{L}_{N\times L}^{(t-1)}-\mathbf{S}_{N\times L}\right\|_F^2. \tag{51}$$

The details of the above process to find the low-rank and the sparsity matrix available in [32] can be summarized in Algorithm 1 as follows.

As described in the above implementation of GoDec, four parameters, m, k, ε, and q, must be determined in advance. Unfortunately, finding appropriate values for these parameters is very challenging and GoDec does not provide any guideline for determining these values and these parameters are generally determined empirically. In a recent work [33], this issue was resolved by using a concept of VD in [8], [34], and [35] and the MX-SVD developed in [36].

Algorithm 1 GoDec

Input: $\mathbf{X}_{N\times L}$ input data matrix
m: the rank of the low rank matrix $\mathbf{L}$
k: SC for sparsity matrix $\mathbf{S}$
ε: predetermined the error
q: power scheme

Output:
$\mathbf{L}_{N\times L}$: Low rank matrix
$\mathbf{S}_{N\times L}$: Sparsity matrix

Initialize: $\mathbf{L}_{N\times L}^{(0)}=\mathbf{X}_{N\times L}$, $\mathbf{S}_{N\times L}^{(0)}=\mathrm{O}_{N\times L}$, $t=0$
While $\|\mathbf{X}_{N\times L}-\mathbf{L}_{N\times L}-\mathbf{S}_{N\times L}^{(t-1)}\|_F^2/\ \|\mathbf{X}_{N\times L}\|_F^2>\varepsilon$ **Do**

$$\begin{aligned}
t &= t+1;\\
\tilde{\mathbf{L}} &= \left[\left(\mathbf{X}_{N\times L}-\mathbf{S}_{N\times L}^{t-1}\right)\left(\mathbf{X}_{N\times L}-\mathbf{S}_{N\times L}^{t-1}\right)^T\right]^{\mathbf{q}}\\
&\quad\times\left(\mathbf{X}_{N\times L}-\mathbf{S}_{N\times L}^{t-1}\right)\\
\mathbf{Y}_1 &= \tilde{\mathbf{L}}\mathbf{A}_1,\quad \mathbf{A}_2=\mathbf{Y}_1;\\
\mathbf{Y}_2 &= \left(\tilde{\mathbf{L}}\right)^T\mathbf{Y}_1=\mathbf{Q}_2\mathbf{R}_2,\quad \mathbf{Y}_1=\tilde{\mathbf{L}}\mathbf{Y}_2=\mathbf{Q}_1\mathbf{R}_1;\\
\mathbf{L}_{N\times L}^{(t)} &= \mathbf{Q}_1\left[\mathbf{R}_1\left(\mathbf{A}_2^T\mathbf{Y}_1\right)^{-1}(\mathbf{R}_2)^T\right]^{1/(2q+1)}\mathbf{Q}_2^T;\\
\mathbf{S}_{N\times L}^{(t)} &= P_\Omega\left(\mathbf{X}_{N\times L}-\mathbf{L}_{N\times L}^{(t)}\right)
\end{aligned} \tag{52}$$

where $P_\Omega(\cdot)$ is the projection of a matrix onto an entry set Ω which will be discussed in detail in Section VI.
End While

By taking advantage of the GoDec-generated LRaSMD model, CEM-AD can operate the data sample $\mathbf{r}$ in the sparse space $\mathbf{S}$ to detect anomalies while inverting $\mathbf{R}$ formed by the BKG space for BS and it is particularly specified by

$$\delta_{\kappa=1}^{\text{CEM-AD}}(\mathbf{r})=\mathbf{r}_{\mathbf{S}}^T\mathbf{R}_{\text{BKG}}^{-1}\mathbf{r}_{\mathbf{S}}. \tag{53}$$

Similarly, OSP-AD can also be implemented by operating data sample $\mathbf{r}$ in sparse space $\mathbf{S}$ to extract anomalies and $P_{\mathbf{U}_{p-1}}^{\perp}$ with $\mathbf{U}_{p-1}$ specified by the BKG space for BKG annihilation and it is specified by

$$\delta_{\kappa=1}^{\text{OSP-AD}}(\mathbf{r})=\mathbf{r}_{\mathbf{S}}^T P_{\text{BKG}}^{\perp}\mathbf{r}_{\mathbf{S}}. \tag{54}$$

The key success in implementing (53) and (54) is how to select $\mathbf{S}$ and BKG.

The major novelty of this section is to implement CEM-AD and OSP-AD in the BKG space and sparse space separately.

VII. Anomaly Space Determined by SC

Since the sparse space $\mathbf{S}$ derived from GoDec is sandwiched between the low rank space $\mathbf{L}$ and noise space $\mathbf{N}$ in (46), $\mathbf{S}$ may contain leakage from the BKG space, $\mathbf{L}$, and a leakage from strong noise in the noise space, $\mathbf{N}$. To resolve this dilemma, GoDec introduced $P_\Omega(\cdot)$ in (52) to determine SC, k in [14] which was determined by $k=j\times N$ in [37] with j being the rank of $\mathbf{S}$. However, this parameter determination is empirical without justification. In what follows, we describe how SC can be determined.

First of all, let's assume that N_r, N_c, and L are the number of rows, number of columns, and number of spectral bands, respectively. Let $\mathbf{r}_{(n_r,n_c)}^{\mathbf{S}}=(r_{(n_r,n_c),1},r_{(n_r,n_c),2},\ldots,r_{(n_r,n_c),L})^T$ be a data sample in $\mathbf{S}$, $\{\mathbf{r}_{(n_r,n_c)}^{\mathbf{S}}=(r_{(n_r,n_c),1},r_{(n_r,n_c),2},\ldots,r_{(n_r,n_c),L})^T\}_{n_r=1,n_c=1}^{N_r,N_c}$, where $r_{(n_r,n_c),l}$ is the lth band value of the data sample $\mathbf{r}_{(n_r,n_c)}^{S}$ at spatial location (n_r,n_c). So, there are a total of N data samples in $\mathbf{S}$, each of which is an L-dimensional data sample vector, $\mathbf{r}_{(n_r,n_c)}=(r_{(n_r,n_c),1},r_{(n_r,n_c),2},\ldots,r_{(n_r,n_c),L})^T$, with $n_r\times n_c=N$.

A. FSC Representation

A natural and intuitive approach to construct a sparse space for $\mathbf{X}_{\text{SC}}$ is to represent each data sample vector $\mathbf{r}_{(n_r,n_c)}^{\mathbf{S}}$ by retaining its j largest spectral component values and setting the remaining spectral component values to zeros. More specifically, for each $\mathbf{r}_{(n_r,n_c)}^{S}$, we define $r_{(n_r,n_c)}^{j}$ as the jth largest value among $\{r_{(n_r,n_c),l}\}_{l=1}^{L}$ in the data sample vector $\mathbf{r}_{(n_r,n_c)}^{S}$, then we can construct a sparse space by

$$r_{(n_r,n_c),l}^{j\text{-FSC}}=\begin{cases} r_{(n_r,n_c),l}, & \text{if } r_{(n_r,n_c),l}\ge r_{(n_r,n_c)}^{j}\\ 0, & \text{if } r_{(n_r,n_c),l}<r_{(n_r,n_c)}^{j}.\end{cases} \tag{55}$$

In this case, we can form $\mathbf{X}_{\text{SC}}$ according to (55) as

$$\mathbf{X}_{j\text{-FSC}}=\left\{\mathbf{r}_{(n_r,n_c)}^{j\text{-FSC}}=\left(r_{(n_r,n_c),1}^{j\text{-FSC}},r_{(n_r,n_c),2}^{j\text{-FSC}},\ldots,r_{(n_r,n_c),L}^{j\text{-FSC}}\right)^T\right\}_{n_r=1,n_c=1}^{N_r,N_c} \tag{56}$$

which consists of j-sparse representations, $\{\mathbf{r}_{(n_r,n_c)}^{j\text{-FSC}}\}_{n_r=1,n_c=1}^{N_r,N_c}$, of $\{\mathbf{r}_{(n_r,n_c)}^{\mathbf{S}}\}_{n_r=1,n_c=1}^{N_r,N_c}$ in $\mathbf{S}$ and each $\mathbf{r}_{(n_r,n_c)}^{j\text{-FSC}}$ has no more than j nonzero spectral component values. So, the resulting

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sparse space $\mathbf{X}_{j\text{-FSC}}$ constructed by $\{\mathbf{r}^{j\text{-FSC}}_{(n_r,n_c)}\}^{N_r,N_c}_{n_r=1,n_c=1}$ is a j-dimensional sparse space where j can be considered as a sparsity of order. As a result, a total of nonzero elements in $\mathbf{X}_{j\text{-FSC}}$ are no greater than $N \times j = k$. The representation (55) can be considered as an alternative sparse representation to the commonly used sparse representation of $\mathbf{r}^{\mathbf{S}}_{(n_r,n_c)}$ constrained by sparsity order j, i.e., $\|\mathbf{r}^{\mathbf{S}}_{(n_r,n_c)}\|_{l_0} \le j$ where $\|\mathbf{x}\|_{l_0}$ is the l_0 norm defined as number of nonzero elements to represent the data sample vector $\mathbf{x}$.

B. VSC Representation

A second approach was SC proposed in [37]. We now rearrange $\{r_{(n_r,n_c),l}\}^{N_r,N_c,L}_{n_r,n_c,l}$ in $\mathbf{S}$ as $\{r_l\}^{N_r\times N_c\times L}_{l=1}$ according to descending order by

$$r_t = r_{(t_r,t_c),l_1} \ge r_s = r_{(s_r,s_c),l_2} \tag{57}$$

Let r_k be the kth largest value among $\{r_t\}^{N_r\times N_c\times L}_{t=1}$. Then, Ω defined by (52) in the GoDec algorithm is the set consisting of the k largest nonzero entries, $\{r_l\}^{k}_{l=1}$, in (57) with $r_l \ne 0$. In this case, some of $r_{(n_r,n_c),l}$ in Ω may belong to the same data sample vector $\mathbf{r}_{(n_r,n_c)}$ in different spectral band components. Accordingly, we can define a VSC space, $\mathbf{X}_{\text{VSC}}$, of $\mathbf{S} = \{\mathbf{r}^{\mathbf{S}}_{(n_r,n_c)}\}^{N_r,N_c}_{n_r,n_c}$ as

$$\mathbf{X}_{\text{VSC}} = \left\{\mathbf{r}^{\text{VSC}}_{(n_r,n_c)} = \left(r^{\text{VSC}}_{(n_r,n_c),1}, r^{\text{VSC}}_{(n_r,n_c),2}, \ldots, r^{\text{VSC}}_{(n_r,n_c),L}\right)^T\right\}^{N_r,N_c}_{n_r=1,n_c=1} \tag{58}$$

where $\mathbf{r}^{\text{VSC}}_{(n_r,n_c)}$ is a sparse representation of $\mathbf{r}^{\mathbf{S}}_{(n_r,n_c)}$ specified by

$$r^{\text{VSC}}_{(n_r,n_c),l} = \begin{cases} r_{(n_r,n_c),l}, & \text{if } r_{(n_r,n_c),l} \in \Omega \\ 0, & \text{if } r_{(n_r,n_c),l} \notin \Omega. \end{cases} \tag{59}$$

Mathematically, let

$$\tau = \min_{r_{(n_r,n_c),l}\in\Omega} r_{(n_r,n_c),l} = r_k \tag{60}$$

be a desired threshold. Then, (51) can be reexpressed as

$$r^{\text{VSC}}_{(n_r,n_c),l} = \begin{cases} r_{(n_r,n_c),l}, & \text{if } r_{(n_r,n_c),l} \ge \tau \\ r_{(n_r,n_c),l} = 0, & \text{if } r_{(n_r,n_c),l} < \tau \end{cases} \tag{61}$$

which implies that $\mathbf{X}_{\text{VSC}}$ in (58) contains only data sample vectors with at least one nonzero spectral component $r_{(n_r,n_c),l} \ge \tau = r_k$. For those data sample vectors not in $\mathbf{X}_{\text{VSC}}$ but in $\mathbf{S}$, denoted by $\mathbf{S} - \mathbf{X}_{\text{SC}}$, they are not considered as potential anomalous data samples. For any data sample vector in $\mathbf{X}_{\text{VSC}}$, $\mathbf{r}^{\text{VSC}}_{(n_r,n_c)}$ is the VSC representation of $\mathbf{r}^{\mathbf{S}}_{(n_r,n_c)}$. If we define that $|A|$ is defined as the number of elements in the set A, then $|\{r_{(n_r,n_c),l}\}^{L}_{l=1} \cap \mathbf{\Omega}| = k$. For example, if $\mathbf{r}^{\text{VSC}}_{(n_r,n_c)}$ has b spectral components, i.e., $\|\mathbf{r}^{\text{VSC}}_{(n_r,n_c)}\|_{l_0} = b$ which are nonzero, then $\{r_{(n_r,n_c),l_i}\}^{b}_{i=1} \subset \mathbf{\Omega}$.

There are two extreme cases for sparse data sample vectors in $\mathbf{S}$.

1) At one extreme, every data sample vector in $\mathbf{S}$ has exactly j nonzero spectral components so that $N \times j = k$. This is the largest possible size of $\mathbf{S}$, i.e., $\mathbf{X}_{\text{SC}} = \mathbf{X}_{j\text{-FSC}} = \mathbf{X}_{\text{VSC}}$. The resulting sparse space constructed by $\mathbf{X}_{\text{SC}}$ is a j-dimensional space where j is an FSC.
2) At the other extreme, there are only $[(k/L)] = [(j \times N)/L]$ data sample vectors in $\mathbf{S}$ which have all L spectral components being nonzero. This is the smallest possible size of $\mathbf{S}$. So, the number of elements in $\mathbf{X}_{j\text{-FSC}}$ or $\mathbf{X}_{\text{VSC}}$ is right between these two extremes, i.e.

$$\left[\frac{j \times N}{L}\right] \le |\mathbf{X}_{j\text{-FSC}}|, \quad |\mathbf{X}_{\text{VSC}}| \le N. \tag{62}$$

It is worth noting that this entire section is new. The key novelty is to interpret SC in terms of sparse representation and further develop two versions of SC, j-FSC and VSC where the value of j is determined by MX-SVD. Interestingly, according to [36], j-FSC and VSC can be interpreted as fixed-length coding and variable length coding, respectively.

VIII. Three-Dimensional ROC Curve-Derived Detection Measures

In AD, one of the important indicators is BS. The commonly used AUC of an ROC curve of P_D versus P_F, $AUC_{(D,F)}$, has been shown not sufficient in [15]. This is because both P_D and P_F are determined by the same cutoff threshold τ. As a result, P_D and P_F are interdependent and cannot stand alone to fully exploit the detection performance. That is, a higher value of P_D also implies a higher value of P_F. In contrast, a lower value of P_D also results in a lower value of P_F. Consequently, $AUC_{(D,F)}$ cannot be used to measure BS. In the literature, we have seen that many works reported very high values of $AUC_{(D,F)}$ for AD. Unfortunately, BS measures have not been introduced for performance evaluation. To address this issue, Chang develops a 3-D ROC analysis to generate a 3-D ROC curve of (P_D,P_F,τ) from which three 2-D ROC curves can also be generated, 2-D ROC curve of (P_D,P_F), 2-D ROC curve of (P_D,τ), and 2-D ROC curve of (P_F,τ). Using AUC of these three 2-D ROC curves, eight detection measures can be derived in [15] and also used to evaluate detection performance from all aspects as follows.

1) Effectiveness of a detector:

$$0 \le \text{AUC}_{(D,F)} \le 1. \tag{63}$$

2) TD, denoted by AUC_{TD} and defined as

$$0 \le \text{AUC}_{(D,\tau)} \le 1 \tag{64}$$

$$0 \le \text{AUC}_{\text{TD}} = \text{AUC}_{(D,F)} + \text{AUC}_{(D,\tau)} \le 2. \tag{65}$$

3) BS measure, denoted by AUC_{BS} and defined as

$$0 \le \text{AUC}_{(F,\tau)} \le 1 \tag{66}$$

$$-1 \le \text{AUC}_{\text{BS}} = \text{AUC}_{(D,F)} - \text{AUC}_{(F,\tau)} \le 1. \tag{67}$$

4) TDBS, denoted by AUC_{TDBS} and defined as

$$-1 \le \text{AUC}_{\text{TDBS}} = \text{AUC}_{(D,\tau)} - \text{AUC}_{(F,\tau)} \le 1. \tag{68}$$

5) ODP, denoted by AUC_{ODP} and defined as

$$-1 \le \text{AUC}_{\text{ODP}} = \text{AUC}_{(D,F)} + \text{AUC}_{(D,\tau)} - \text{AUC}_{(F,\tau)} \le 2. \tag{69}$$

6) SNPR, denoted by AUC_{SNPR} and defined as

$$0 \le \text{AUC}_{\text{SNPR}} = \frac{\text{AUC}_{(D,\tau)}}{\text{AUC}_{(F,\tau)}}. \tag{70}$$

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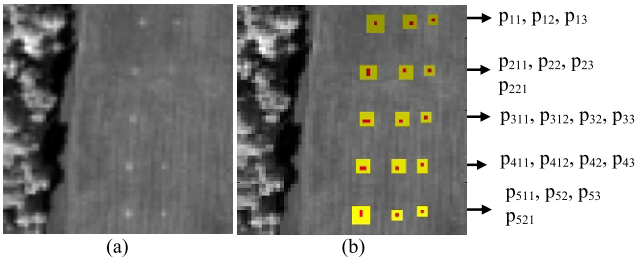


Fig. 2. (a) HYDICE15-panel scene which contains 15 panels and (b) ground-truth map of spatial locations of 19 *R* panel pixels.

It should be noted that except the case of (66) where a smaller value of $\mathrm{AUC}_{(\mathrm{F},\tau)}$ indicates a better BS, all the other seven detection measures perform better if their corresponding values are higher.

IX. Real Hyperspectral Image Experiments

A. Hyperspectral Images Used for Experiments

Three hyperspectral real image scenes were used for experiments.

1) HYDICE 15-Panel Scene: An airborne Hyperspectral Digital Imagery Collection Experiment (HYDICE) scene is shown in Fig. 2. It was collected in August 1995 from a flight altitude of 10 000 ft with the ground sampling distance approximately 1.56 m. This scene has been studied extensively by many reports such as [3], [8], [26]. A total of 169 bands were used for the experiments with low signal/high noise bands: bands 1–3 and bands 202–210; and water vapor absorption bands: bands 101–112 and bands 137–153 removed. There are 15 square panels in Fig. 2(a) with three different sizes, 3 m $\times$ 3 m, 2 m $\times$ 2 m, and 1 m $\times$ 1 m, respectively. Due to the ground sampling distance approximately 1.56 m, each panel in the first column except the first row contains two panel pixels highlighted by red, p_{211}, p_{221} in row 2, p_{311}, p_{312} in row 3, p_{411}, p_{412} in row 4, and p_{511}, p_{521} in row 5, as shown in Fig. 2. All the remaining 11 panels in Fig. 2 contain one single panel pixel for each panel also highlighted by red, p_{11}, p_{12}, p_{13} in row 1, p_{22}, p_{23} in row 2, p_{32}, p_{33} in row 3, p_{42}, p_{43} in row 4, and p_{52}, p_{53} in row 5. Therefore, there are a total of 19 red (R) panel pixels. Fig. 2(b) shows their precise spatial locations with the pixels in yellow (Y pixels) indicating panel pixels mixed with the BKG.

The best part of this scene is that it contains pure pixel panels in the first column, pure and mixed pixel panels in the second column, and subpixel panels in the third column. These 19 *R* panels offer the ground truth to evaluate how effectively an AD finds the targets of various types.

2) HYDICE Urban Scene: A second dataset is another HYDICE image with pseud-color shown in Fig. 3(a).

It is an urban scene and comprised 210 spectral bands with 174 bands being used for experiments after the noise and water absorption bands had been removed. A region with a size of 80 $\times$ 100 pixels located at the upper right of the scene was selected as the test image shown in Fig. 3(b) along with the ground-truth map shown in Fig. 3(c) where twenty-one pixels were identified as anomalies, which were cars and roofs, because they had spectra that differ from BKG. This scene

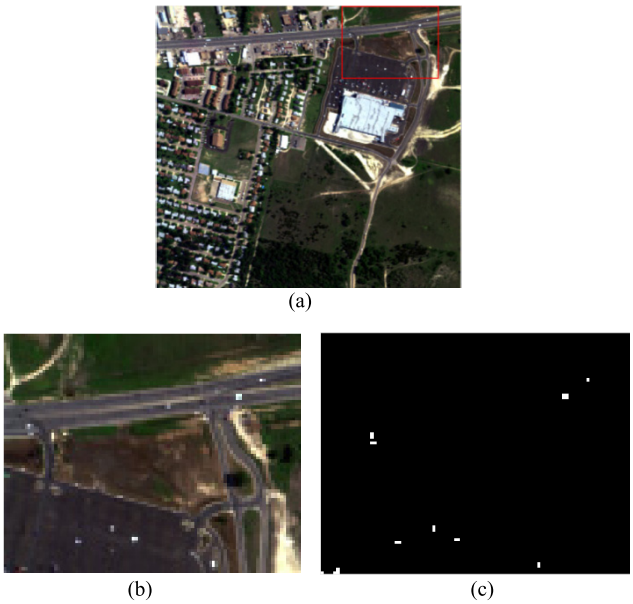


Fig. 3. HYDICE urban scene (a) pseudo-color image of the whole scene, (b) pseudo-color image of the selected area, and (c) ground-truth map.

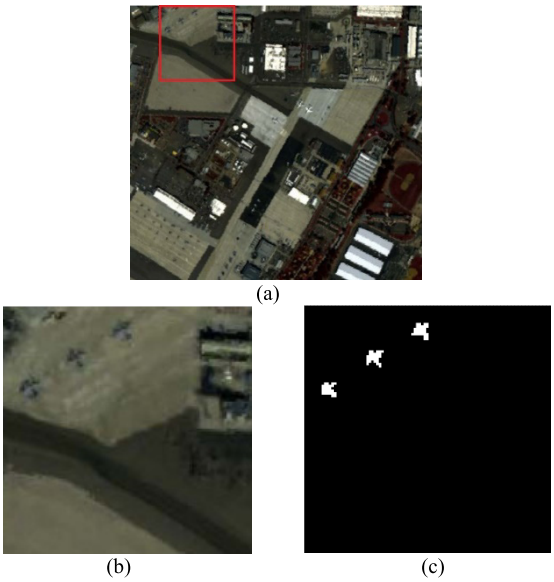


Fig. 4. San Diego Airport scene (a) pseudo-color image of the whole scene, (b) pseudo-color image of the selected area, and (c) ground-truth map.

provides many anomalous pixels which are generally invisible by visual inspection. So, detection of these target pixels can be used to evaluate AD performance.

3) San Diego Airport Data: A third dataset used for experiments was also an Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) acquired scene, which is a San Diego airport scene in CA, USA with pseudo-color shown in Fig. 4(a) [38]. It has a size of 400 $\times$ 400 pixels with a 3.5-m spatial resolution and 224 spectral channels in wavelengths ranging from 370 to 2510 nm. After removing the bands that correspond to the water absorption regions, low SNR, and bad bands (1–6, 33–35, 97, 107–113, 153–166, and 221–224), 189 available bands of the data were retained in the experiments. An area of 100 $\times$ 100 pixels at the upper left corner of the scene was selected as the test image shown

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TABLE I

p AND m ESTIMATED FOR THREE DATASETS

	p	j by MX-SVD
HYDICE 15-panel scene	13 by NWHFC	6
HYDICE urban scene	11 by HFC	4
San Diego airport	10 by HFC	8

TABLE II

PARAMETERS FOR $\delta^{\text{CRD}}(\mathbf{r})$

Data	w_{in}	w_{out}	λ
HYDICE 15-panel	7	11	10^{-4}
HYDICE Urban	9	11	10^{-4}
San Diego Airport	9	15	10^{-4}

TABLE III

COMPUTER ENVIRONMENT

Environment Parameter	Configuration
OS Name:	Microsoft Windows 10 Home
OS Version:	10.0.19043 N/A Build 19043
OS Manufacturer:	Microsoft Corporation
System Type:	x64-based PC
Processor(s):	Intel(R) Core(TM) i7-8750H CPU @ 2.20GHz 2.21 GHz
Total Physical Memory:	32,624 MB
Graphics Card:	GeForce GTX 1060 Graphics Cards

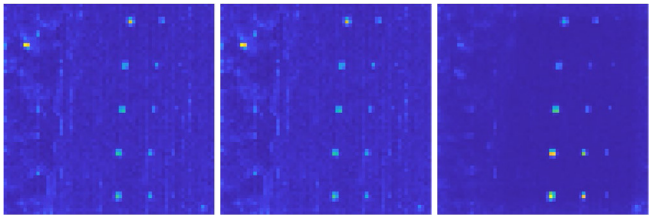
in Fig. 4(b) along with its ground-truth map shown in Fig. 4(c). It is an urban scene in which the main background materials are roof, shadow, and grass. There are three airplanes in the image, which consist of 58 pixels and account for 0.33% of the test image, as shown in Fig. 3(b). This scene is less interesting than the HYDICE 15-panel scene in Fig. 2 and HYDICE urban scene in Fig. 3 because only available ground truth is three airplanes which are very large visible targets. In reality, many subtle unknown target pixels are actually present but cannot be identified by visual inspection. So, technically speaking, this data scene is more suitable for target detection rather than AD. This will become clear when the experiments conducted in Section IX-B will show quite different detection results between this scene and the other two HYDICE scenes. Since this scene has been widely used for AD in the literature, it is included for the reason that we can make comparison against many currently developed ADs.

4) Parameters Setting: Table I calculated the value of p estimated by HFC and NWHFC methods with the false alarm probability, $P_F = 10^{-3}$, and the value of j estimated by MX-SVD for all the three scenes to be used for GoDec. Table II tabulates the parameters used by CRD in the experiments and Table III provides the details of the computer environment used for experiments.

Table III provides the computer environment used for the experiments.

B. Experimental Results and Discussions

In the following experiments, five types of ADs were tested and compared with specific reasons. Included are three types of ADs (iii–v) using the LRaSMD model produced by GoDec.



(a) (b) (c)

Fig. 5. Detection maps of (a) $\delta^{\text{RX-AD}}(\mathbf{r})$, (b) $\delta^{\text{R-AD}}(\mathbf{r})$, and (c) $\delta^{\text{CRD}}(\mathbf{r})$.

1) RX-AD($\delta^{\text{RX-AD}}(\mathbf{r})$) [4] and R-AD($\delta^{\text{R-AD}}(\mathbf{r})$) [31]. These are two classic ADs used for benchmark comparison.
2) CRD($\delta^{\text{CRD}}(\mathbf{r})$)
This work is a representative of window-based ADs, which has been shown to perform better than many existing ADs [39].
3) RX-AD operating in $\mathbf{X}_{j\text{-FSC}}$ and $\mathbf{X}_{\text{VSC}}$ produced by GoDec via (43) and (44) which has shown to perform better than RPCA [19], CRD, and LRaSMD-based Mahalanobis distance AD (LSMAD) [37]

$$\delta^{\text{RX-AD}}_{\text{GoDec}}(\mathbf{r}) = (\mathbf{r}_{\mathbf{S}} - \boldsymbol{\mu}_{\text{BKG}})^T \mathbf{K}^{-1}_{\text{BKG}}(\mathbf{r}_{\mathbf{S}} - \boldsymbol{\mu}_{\text{BKG}})$$

$\delta^{\text{RX-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\mathbf{S}_j})$, $\delta^{\text{RX-AD}}_{\mathbf{L}_{m_j}}(\mathbf{r}_{\text{FSC}_j})$, $\delta^{\text{RX-AD}}_{\mathbf{L}_{m_j}}(\mathbf{r}_{\text{VSC}_j})$, $\delta^{\text{RX-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$.
4) CEM-AD in (28) operating in $\mathbf{X}_{j\text{-FSC}}$ and $\mathbf{X}_{\text{VSC}}$ produced by GoDec via (43) and (44)

$$\delta^{\text{CEM-AD}}_{\kappa=1}(\mathbf{r}) = \mathbf{r}^T_{\mathbf{S}} \mathbf{R}^{-1}_{\text{BKG}} \mathbf{r}_{\mathbf{S}}$$

$\delta^{\text{CEM-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\mathbf{S}_j})$, $\delta^{\text{CEM-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\text{FSC}_j})$, $\delta^{\text{CEM-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\text{VSC}_j})$, $\delta^{\text{CEM-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$.
5) OSP-AD [11] in (37) operating in $\mathbf{X}_{j\text{-FSC}}$ and $\mathbf{X}_{\text{VSC}}$ produced by GoDec via (43) and (44)

$$\delta^{\text{OSP-AD}}_{\kappa=1}(\mathbf{r}) = \mathbf{r}^T_{\mathbf{S}} P^{\perp}_{\text{BKG}} \mathbf{r}_{\mathbf{S}}$$

$\delta^{\text{OSP-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\mathbf{S}_j})$, $\delta^{\text{OSP-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\text{FSC}_j})$, $\delta^{\text{OSP-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\text{VSC}_j})$, $\delta^{\text{OSP-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$.

Since the discussions of performance among RX-AD, CRD, RPCA, and LSMAD were available in [11] and [33], those who are interested in these detectors can find details in these references.

1) HYDICE 15-Panel Data: Figs. 5–8 show the detection maps produced by the above four types of ADs. By visual inspection, $\delta^{\text{RX-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$ and $\delta^{\text{CEM-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$ were the best in terms of detecting all the five subpixel panels in the third column, while OSP-AD was the worst. However, $\delta^{\text{CEM-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$ seemed to do better than $\delta^{\text{RX-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$ in the sense of its better BS.

To further conduct a quantitative analysis, the eight 3-D ROC curve-derived detection measures in Section VII were calculated from Figs. 5–8 and tabulated in Tables IV–VI where the best results are boldfaced and computing running times are also documented. Interestingly, according to the visual inspection, $\delta^{\text{CEM-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$ performed slightly better than $\delta^{\text{RX-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$. However, the results in Tables IV–VI showed that $\delta^{\text{RX-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$ indeed performed slightly

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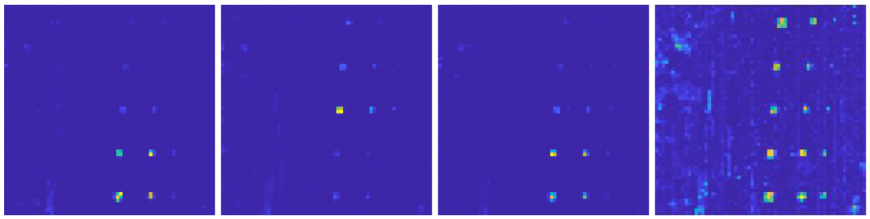


Fig. 6. Detection maps of RX-AD using SC with $p = 13$, $m = 7$, and $j = 6$. (d) $\delta^{\text{RX-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\mathbf{S}_j})$. (e) $\delta^{\text{RX-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\text{FSC}_j})$. (f) $\delta^{\text{RX-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\text{VSC}_j})$. (g) $\delta^{\text{RX-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$.

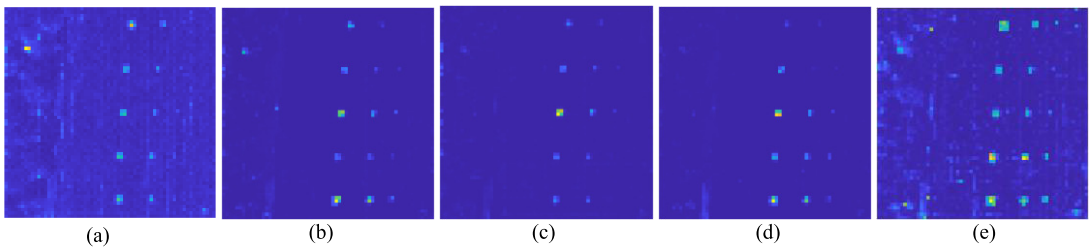


Fig. 7. Detection maps of CEM-AD using SC with $p = 13$, $m = 7$, and $j = 6$. (a) $\delta^{\text{CEM-AD}}(\mathbf{r})$. (b) $\delta^{\text{CEM-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\mathbf{S}_j})$. (c) $\delta^{\text{CEM-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\text{FSC}_j})$. (d) $\delta^{\text{CEM-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\text{VSC}_j})$. (e) $\delta^{\text{CEM-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$.

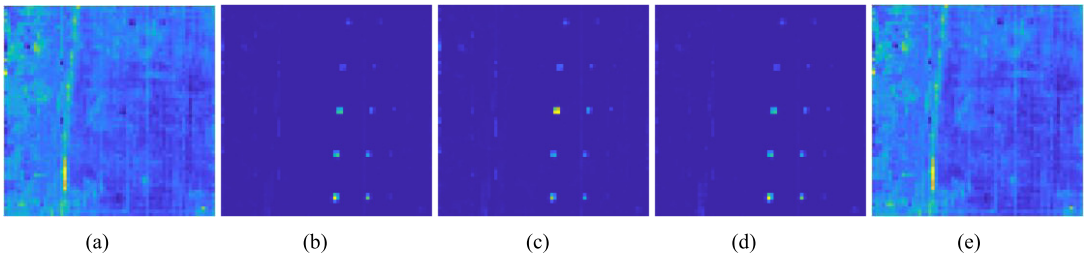


Fig. 8. Detection maps of OSP-AD using SC with $p = 13$, $m = 7$, and $j = 6$. (a) $\delta^{\text{OSP-AD}}(\mathbf{r})$. (b) $\delta^{\text{OSP-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\mathbf{S}_j})$. (c) $\delta^{\text{OSP-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\text{FSC}_j})$. (d) $\delta^{\text{OSP-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\text{VSC}_j})$. (e) $\delta^{\text{OSP-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$.

TABLE IV
AUC VALUES CALCULATED FROM THE THREE 2-D ROC CURVES FOR FIVE ROWS DETECTION RESULTS

detector	$\text{AUC}_{(\text{D,F})}$	$\text{AUC}_{(\text{D},\tau)}$	$\text{AUC}_{(\text{F},\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{SNPR}	AUC_{TDBS}	AUC_{ODP}	Time(s)
$\delta^{\text{RX-AD}}(\mathbf{r})$	0.9898	0.3574	0.0430	1.3472	0.9466	8.3205	0.3145	1.3042	0.242436
$\delta^{\text{R-AD}}(\mathbf{r})$	0.9900	0.3584	0.0435	1.3484	0.9462	8.2341	0.3149	1.3049	**0.159342**.
$\delta^{\text{CRD}}(\mathbf{r})$	0.9948	0.4319	0.0181	1.4267	0.9764	23.8392	0.4137	1.4085	5.873875
$\delta^{\text{RX-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\mathbf{S}_j})$	0.9894	0.2610	0.0029	1.2504	0.9859	89.6188	0.2580	1.2474	2.697250
$\delta^{\text{RX-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\text{FSC}_j})$	0.9403	0.1942	0.0032	1.1345	0.9338	60.8181	0.1910	1.1313	4.989418
$\delta^{\text{RX-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\text{VSC}_j})$	0.9927	0.2871	**0.0021**	1.2798	**0.9902**	**138.65**	0.2851	1.2778	2.668299
$\delta^{\text{RX-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$	**0.9952**	**0.6697**	0.0320	**1.6648**	0.9629	20.9578	**0.6377**	**1.6329**	2.659375

better than $\delta^{\text{CEM-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$ in terms of TD and overall detection performance. However, $\delta^{\text{CEM-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$ seemed to do better than $\delta^{\text{RX-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$ in the sense of its better BS. As for computing time, CRD required the highest computational time as opposed to R-AD which requires the least computational time.

2) HYDICE Urban Data: Figs. 9–12 show the detection maps produced by the above four types of ADs. By visual inspection, both $\delta^{\text{RX-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$, and $\delta^{\text{CEM-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$ were the best and OSP-AD was the worst in terms of detecting anomalies where $\delta^{\text{CEM-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$ seemed to do better than $\delta^{\text{RX-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$ in the sense of its better BS.

Similar to Tables IV–VI, Tables VII–IX calculated the eight 3-D ROC curve-derived detection measures from Figs. 9–12 and the computing running times of all test ADs are also documented where the best results are boldfaced. It is clear

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TABLE V
AUC VALUES CALCULATED FROM THE THREE 2-D ROC CURVES FOR FIVE ROWS DETECTION RESULTS

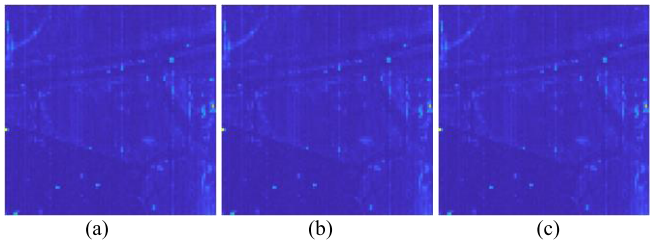
detector	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{SNPR}	AUC_{TDBS}	AUC_{ODP}	Times (s)
$\delta^{\text{CEM-AD}}(\mathbf{r})$	0.9900	0.3584	0.0435	1.3484	0.9462	8.2341	0.3149	1.3049	**0.275658**
$\delta^{\text{CEM-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\mathbf{S}_j})$	0.9885	0.2488	**0.0021**	1.2373	0.9858	120.8411	0.2468	1.2353	2.721553
$\delta^{\text{CEM-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\text{FSC}_j})$	0.9689	0.2060	0.0034	1.1750	0.9638	59.9586	0.2026	1.1715	4.897384
$\delta^{\text{CEM-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\text{VSC}_j})$	0.9897	0.3366	0.0028	1.3263	**0.9864**	**121.9857**	0.3338	1.3236	2.781287
$\delta^{\text{CEM-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$	**0.9949**	**0.6261**	0.0277	**1.6210**	0.9669	22.5765	**0.5984**	**1.5933**	2.973819

TABLE VI
AUC VALUES CALCULATED FROM THE THREE 2-D ROC CURVES FOR FIVE ROWS DETECTION RESULTS

detector	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{SNPR}	AUC_{TDBS}	AUC_{ODP}	Time (s)
$\delta^{\text{OSP-AD}}(\mathbf{r})$	0.2280	0.1255	0.2351	0.3535	-0.0072	0.5340	-0.1095	0.1184	**0.329725**
$\delta^{\text{OSP-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\mathbf{S}_j})$	**0.9907**	0.3041	**0.0029**	1.2948	**0.9872**	103.5765	0.3012	1.2918	2.715872
$\delta^{\text{OSP-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\text{FSC}_j})$	0.9727	0.2335	0.0053	1.2063	0.9659	44.1294	0.2283	1.2010	4.975631
$\delta^{\text{OSP-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\text{VSC}_j})$	0.9905	**0.3088**	0.0030	**1.2993**	0.9870	**103.6175**	**0.3058**	**1.2963**	2.802578

TABLE VII
AUC VALUES CALCULATED FROM THE THREE 2-D ROC CURVES FOR HYDICE URBAN DETECTION RESULTS

detector	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{SNPR}	AUC_{TDBS}	AUC_{ODP}	Time(s)
$\delta^{\text{RX-AD}}(\mathbf{r})$	0.9872	0.2575	0.0361	1.2448	0.9511	7.1406	0.2215	1.2087	0.346452
$\delta^{\text{R-AD}}(\mathbf{r})$	0.9872	0.2549	0.0359	1.2421	0.9511	7.0965	0.2190	1.2062	**0.184774**
$\delta^{\text{CRD}}(\mathbf{r})$	**0.9964**	0.4030	0.0438	1.3994	0.9524	9.1900	0.3591	1.3556	5.424751
$\delta^{\text{RX-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\mathbf{S}_j})$	0.9619	0.1109	**0.0036**	1.0729	0.9583	31.1580	0.1074	1.0693	5.755268
$\delta^{\text{RX-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\text{FSC}_j})$	0.9546	0.3402	0.0240	1.2948	0.9283	14.1610	0.3162	1.2708	9.886829
$\delta^{\text{RX-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\text{VSC}_j})$	0.9568	0.1011	0.0041	1.0579	0.9525	24.5337	0.0970	1.0538	5.700472 s
$\delta^{\text{RX-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$	0.9861	**0.6555**	0.0192	**1.6416**	**0.9668**	**34.1216**	**0.6363**	**1.6224**	5.808477



(a) (b) (c)

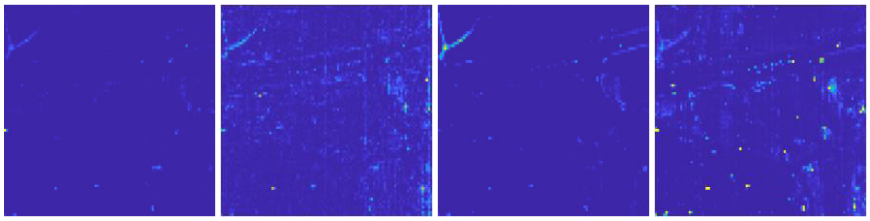
Fig. 9. Detection maps of RX-AD, R-AD, and CRD. (a) $\delta^{\text{RX-AD}}(\mathbf{r})$. (b) $\delta^{\text{R-AD}}(\mathbf{r})$. (c) $\delta^{\text{CRD}}(\mathbf{r})$.

that $\delta^{\text{CEM-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$ was the best in 5 of 8 detection measures. However, if we used only $AUC_{(D,F)}$ value as the criterion to measure detection performance, CRD was the best, but its other seven detection measures were actually not good, particularly, BS.

The above experiments provided hard evidence that solely relying on $AUC_{(D,F)}$ value is misleading. As for computing time, OSP-AD required the more computational times than other ADs.

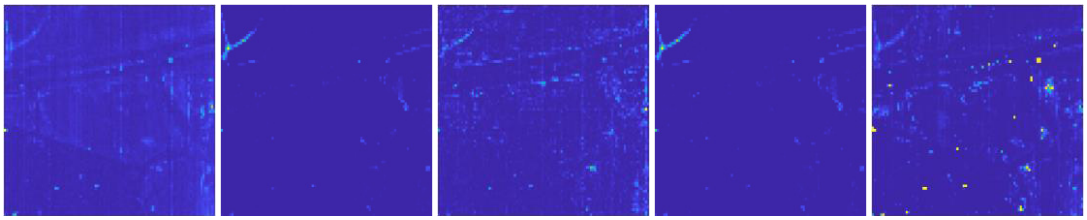
3) San Diego Airport Data: Figs. 13–16 show the detection maps produced by the above four types of ADs. By visual inspection, there are some interesting observations which were different from that we observed for the HYDICE 15-panel and HYDICE urban datasets. It seemed that $\delta^{\text{OSP-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\mathbf{S}_j})$, $\delta^{\text{OSP-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\text{FSC}_j})$, $\delta^{\text{OSP-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\text{VSC}_j})$, and $\delta^{\text{OSP-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$ in Fig. 16 performed better than CEM-AD in Fig. 15, RX/R-AD in Fig. 14, and RX/R-AD in Fig. 13 in terms of detecting three airplanes. However, CRD, $\delta^{\text{RX-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\text{VSC}_j})$, and $\delta^{\text{CEM-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$ also performed comparably.

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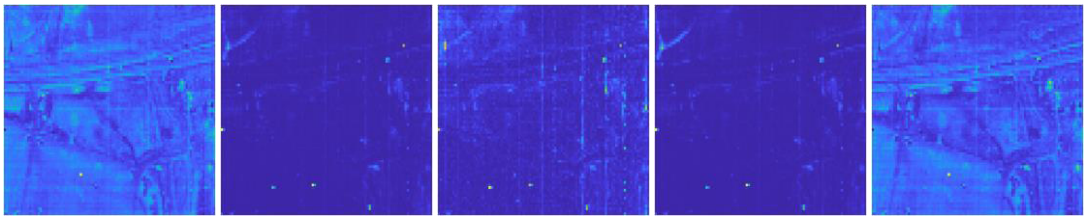
(d) (e) (f) (g)

Fig. 10. Detection maps of RX-AD using SC with $p = 11$, $m = 7$, and $j = 4$. (d) $\delta^{\text{RX-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\mathbf{S}_j})$. (e) $\delta^{\text{RX-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\text{FSC}_j})$. (f) $\delta^{\text{RX-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\text{VSC}_j})$. (g) $\delta^{\text{RX-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$.



(a) (b) (c) (d) (e)

Fig. 11. Detection maps of CEM-AD using SC with $p = 11$, $m = 7$, and $j = 4$. (a) $\delta^{\text{CEM-AD}}(\mathbf{r})$. (b) $\delta^{\text{CEM-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\mathbf{S}_j})$. (c) $\delta^{\text{CEM-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\text{FSC}_j})$. (d) $\delta^{\text{CEM-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\text{VSC}_j})$. (e) $\delta^{\text{CEM-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$.



(a) (b) (c) (d) (e)

Fig. 12. Detection maps of OSP-AD using SC with $p = 11$, $m = 7$, and $j = 4$. (a) $\delta^{\text{OSP-AD}}(\mathbf{r})$. (b) $\delta^{\text{OSP-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\mathbf{S}_j})$. (c) $\delta^{\text{OSP-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\text{FSC}_j})$. (d) $\delta^{\text{OSP-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\text{VSC}_j})$. (e) $\delta^{\text{OSP-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$.

TABLE VIII

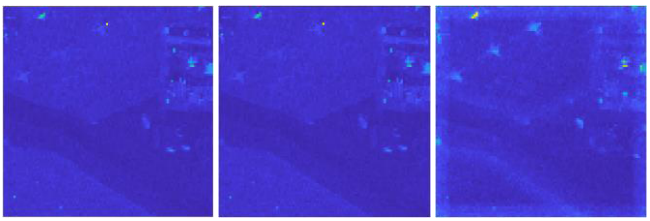
AUC VALUES CALCULATED FROM THE THREE 2-D ROC CURVES FOR HYDICE URBAN DETECTION RESULTS

detector	$\text{AUC}_{(D,F)}$	$\text{AUC}_{(D,\tau)}$	$\text{AUC}_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{SNPR}	AUC_{TDBS}	AUC_{ODP}	Times (s)
$\delta^{\text{CEM-AD}}(\mathbf{r})$	**0.9872**	0.2549	0.0359	1.2421	0.9511	7.0965	0.2190	1.2062	**0.262658**
$\delta^{\text{CEM-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\mathbf{S}_j})$	0.9573	0.0505	**0.0030**	1.0078	0.9542	16.7658	0.0475	1.0048	5.711293
$\delta^{\text{CEM-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\text{FSC}_j})$	0.8894	0.1638	0.0226	1.0532	0.8667	7.2560	0.1412	1.0306	9.866633
$\delta^{\text{CEM-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\text{VSC}_j})$	0.9622	0.0778	0.0034	1.0400	0.9587	22.9137	0.0744	1.0366	5.643523
$\delta^{\text{CEM-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$	0.9818	**0.6771**	0.0185	**1.6589**	**0.9631**	**36.5176**	**0.6586**	**1.6404**	5.686779

To further evaluate visual assessment of the comparable ADs quantitatively, the eight 3-D ROC curve-derived detection measures were calculated from Figs. 13–16 and tabulated in Tables X–XII and the computing running times of all test ADs are also documented where the best results are bold-faced. Speaking on overall performance, $\delta^{\text{OSP-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\mathbf{S}_j})$ seemed the best followed by $\delta^{\text{OSP-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\text{FSC}_j})$ and $\delta^{\text{CEM-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$. These results also confirmed the visual inspection.

C. Comparative Analysis

In Sections IX-B1, IX-B2, and IX-B3, experiments were conducted for five different types of ADs using three different data scenes. We selected the overall best performance of each type, i.e., $\delta^{\text{RX-AD}}(\mathbf{r})$, $\delta^{\text{CRD}}(\mathbf{r})$, $\delta^{\text{RX-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$, $\delta^{\text{CEM-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$, and $\delta^{\text{OSP-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\text{VSC}_j})$



(a) (b) (c)

Fig. 13. Detection maps of RX-AD, R-AD, and CRD. (a) $\delta^{\text{RX-AD}}(\mathbf{r})$. (b) $\delta^{\text{R-AD}}(\mathbf{r})$. (c) $\delta^{\text{CRD}}(\mathbf{r})$.

and showed their detection maps for the HYDICE 15-panel data scene in Fig. 17. By visual inspection, it seemed that $\delta^{\text{CEM-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$ produced the best results among all five types of ADs followed by $\delta^{\text{RX-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$ as the second

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TABLE IX
AUC VALUES CALCULATED FROM THE THREE 2-D ROC CURVES FOR HYDICE URBAN DETECTION RESULTS

detector	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{SNPR}	AUC_{TDBS}	AUC_{ODP}	Time (s)
$\delta^{\text{OSP-AD}}(\mathbf{r})$	0.5639	0.2602	0.1937	0.8240	0.3484	1.3434	0.0665	0.6304	**0.428866**
$\delta^{\text{OSP-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\mathbf{S}_j})$	0.9641	0.3076	0.0136	1.2717	0.9504	**22.5998**	0.2939	1.2581	5.821708
$\delta^{\text{OSP-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\text{FSC}_j})$	0.9584	**0.3747**	0.0506	**1.3332**	0.9058	7.4079	**0.3242**	1.2826	9.820838
$\delta^{\text{OSP-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\text{VSC}_j})$	**0.9842**	0.2952	**0.0135**	1.2793	**0.9706**	21.9462	0.2817	**1.2659**	5.731333
$\delta^{\text{OSP-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$	0.7116	0.2903	0.1671	1.0020	0.5302	1.7378	0.1233	0.8349	5.797485



(d) (e) (f) (g)

Fig. 14. Detection maps of RX-AD using SC with $p = 10$, $m = 2$, and $j = 8$. (d) $\delta^{\text{RX-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\mathbf{S}_j})$. (e) $\delta^{\text{RX-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\text{FSC}_j})$. (f) $\delta^{\text{RX-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\text{VSC}_j})$. (g) $\delta^{\text{RX-AD}}_{\mathbf{L}_m+\mathbf{S}_j}(\mathbf{r}_{\mathbf{L}_m+\mathbf{S}_j})$.



(a) (b) (c) (d) (e)

Fig. 15. Detection maps of CEM-AD using SC with $p = 10$, $m = 2$, and $j = 8$. (a) $\delta^{\text{CEM-AD}}(\mathbf{r})$. (b) $\delta^{\text{CEM-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\mathbf{S}_j})$. (c) $\delta^{\text{CEM-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\text{FSC}_j})$. (d) $\delta^{\text{CEM-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\text{VSC}_j})$. (e) $\delta^{\text{CEM-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$.



(a) (b) (c) (d) (e)

Fig. 16. Detection maps of OSP-AD using SC with $p = 10$, $m = 2$, and $j = 8$. (a) $\delta^{\text{OSP-AD}}(\mathbf{r})$. (b) $\delta^{\text{OSP-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\mathbf{S}_j})$. (c) $\delta^{\text{OSP-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\text{FSC}_j})$. (d) $\delta^{\text{OSP-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\text{VSC}_j})$. (e) $\delta^{\text{OSP-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$.

best since both of them detected subpixel target panels in the third column. However, if we further tabulate their detection results by the eight 3-D ROC curve-derived detection measures in Table XIII where the best results among these five ADs are boldfaced and their computing running times are also documented for reference. The quantitative results showed that $\delta^{\text{RX-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$ performed slightly better than $\delta^{\text{CEM-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$ with only very small differences. The most interesting observation is that $\delta^{\text{OSP-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\text{VSC}_j})$ did very well in BS as confirmed by visual inspection of its detection map in Fig. 17(a)–(e).

For the HYDICE urban scene, we also selected the overall best performance of each type i.e., $\delta^{\text{RX-AD}}(\mathbf{r})$, $\delta^{\text{CRD}}(\mathbf{r})$, $\delta^{\text{RX-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$, $\delta^{\text{CEM-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$, and $\delta^{\text{OSP-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\text{VSC}_j})$, which happened to exactly the same ADs selected for the HYDCE 15-panel data. Fig. 18(a)–(e) shows their detection maps. Once again, both $\delta^{\text{RX-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$ and $\delta^{\text{CEM-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$ were the best ADs among all five types of ADs by visual inspection. To further observe quantitative results, Table XIV calculated the AUC values of the eight detection measures where the best results among these five ADs are boldfaced and their computing running times are also

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TABLE X

AUC VALUES CALCULATED FROM THE THREE 2-D ROC CURVES FOR SAN DIEGO AIRPORT DETECTION RESULTS

detector	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{SNPR}	AUC_{TDBS}	AUC_{ODP}	Time(s)
$\delta^{\text{RX-AD}}(\mathbf{r})$	0.8334	0.0910	0.0453	0.9244	0.7861	2.0101	0.0457	0.8791	0.364277.
$\delta^{\text{R-AD}}(\mathbf{r})$	0.8291	0.0940	0.0501	0.9231	0.7770	1.8778	0.0439	0.8730	**0.198738**
$\delta^{\text{CRD}}(\mathbf{r})$	0.9132	0.1804	0.0684	1.0936	0.8447	2.6371	0.1120	1.0252	43.912223
$\delta^{\text{RX-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\mathbf{S}_j})$	0.8220	0.1205	**0.0049**	0.9426	0.8170	24.3840	0.1156	0.9376	8.217696
$\delta^{\text{RX-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\text{FSC}_j})$	0.9034	0.0576	0.0098	0.9610	0.8935	5.8655	0.0478	0.9512	12.718195
$\delta^{\text{RX-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\text{VSC}_j})$	0.8354	0.1312	**0.0049**	0.9666	0.8304	**26.7800**	**0.1263**	0.9617	8.240556
$\delta^{\text{RX-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$	**0.9336**	**0.1332**	0.0181	**1.0668**	**0.9154**	7.3383	0.1150	**1.0487**	8.165311

TABLE XI

AUC VALUES CALCULATED FROM THE THREE 2-D ROC CURVES FOR SAN DIEGO AIRPORT DETECTION RESULTS

detector	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{SNPR}	AUC_{TDBS}	AUC_{ODP}	Times (s)
$\delta^{\text{CEM-AD}}(\mathbf{r})$	0.8291	0.0940	0.0501	0.9231	0.7770	1.8778	0.0439	0.8730	**0.261189**
$\delta^{\text{CEM-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\mathbf{S}_j})$	0.9309	0.0463	**0.0026**	0.9772	**0.9282**	17.8788	0.0437	0.9746	8.125776
$\delta^{\text{CEM-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\text{FSC}_j})$	0.8890	0.0514	0.0100	0.9404	0.8789	5.1190	0.0413	0.9304	12.350603
$\delta^{\text{CEM-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\text{VSC}_j})$	0.9323	0.1087	0.0047	1.0411	0.9276	**23.3098**	0.1041	1.0364	8.226176
$\delta^{\text{CEM-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$	**0.9347**	**0.1809**	0.0188	**1.1156**	0.9158	9.6212	**0.1621**	**1.0968**	8.046591

TABLE XII

AUC VALUES CALCULATED FROM THE THREE 2-D ROC CURVES FOR SAN DIEGO AIRPORT DETECTION RESULTS

detector	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{SNPR}	AUC_{TDBS}	AUC_{ODP}	Time (s)
$\delta^{\text{OSP-AD}}(\mathbf{r})$	0.2397	0.1361	0.2387	0.3758	0.0011	0.5702	-0.1026	0.1372	0.447020
$\delta^{\text{OSP-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\mathbf{S}_j})$	**0.9607**	0.1961	0.0079	1.1569	**0.9528**	**24.9255**	0.1883	1.1490	8.078681.
$\delta^{\text{OSP-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\text{FSC}_j})$	0.9600	**0.2785**	0.0282	**1.2385**	0.9317	9.8857	**0.2503**	**1.2103**	12.233089
$\delta^{\text{OSP-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\text{VSC}_j})$	0.9462	0.1145	**0.0069**	1.0606	0.9391	16.4796	0.1075	1.0537	8.270531
$\delta^{\text{OSP-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$	0.8912	0.2139	0.0637	1.1051	0.8274	3.3564	0.1502	1.0414	8.068384

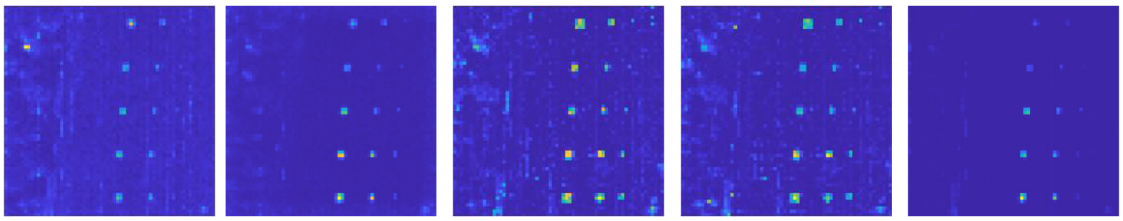


Fig. 17. Detection maps of five types of best representative ADs for HYDICE 15-panel data, $\delta^{\text{RX-AD}}(\mathbf{r})$, $\delta^{\text{CRD}}(\mathbf{r})$, $\delta^{\text{RX-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$, $\delta^{\text{CEM-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$, $\delta^{\text{OSP-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\text{VSC}_j})$.

documented. Overall speaking, $\delta^{\text{CEM-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$ was the best and performed better than $\delta^{\text{RX-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$ with appreciable differences. Interestingly, unlike the HYDICE 15-panel data, $\delta^{\text{CEM-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$ could do as well as $\delta^{\text{OSP-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\text{VSC}_j})$ did in BS.

The San Diego airport scene is quite different from the two HYDICE data scenes discussed above. Since there are many anomalies such as roofs which cannot be identified by ground truth, the only targets of interest in the scene are the three airplanes which are very large and visible.

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TABLE XIII

AUC VALUES CALCULATED FROM THE THREE 2-D ROC CURVES FOR FIVE ROWS DETECTION RESULTS IN FIG. 17

detector	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{SNPR}	AUC_{TDBS}	AUC_{ODP}	Time (s)
$\delta^{\text{RX-AD}}(\mathbf{r})$	0.9898	0.3574	0.0430	1.3472	0.9466	8.3205	0.3145	1.3042	**0.242436**
$\delta^{\text{CRD}}(\mathbf{r})$	0.9948	0.4319	0.0181	1.4267	0.9764	23.8392	0.4137	1.4085	5.873875
$\delta^{\text{RX-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$	**0.9952**	**0.6697**	0.0320	**1.6648**	0.9629	20.9578	**0.6377**	**1.6329**	2.659375
$\delta^{\text{CEM-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$	0.9949	0.6261	0.0277	1.6210	0.9669	22.5765	0.5984	1.5933	2.973819
$\delta^{\text{OSP-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\text{VSC}_j})$	0.9905	0.3088	**0.0030**	1.2993	**0.9870**	**103.6175**	0.3058	1.2963	2.802578

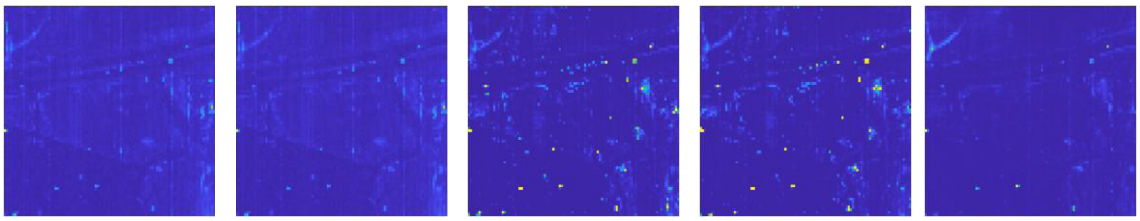


Fig. 18. Detection maps of five types of best representative ADs for HYDICE urban, $\delta^{\text{RX-AD}}(\mathbf{r})$, $\delta^{\text{CRD}}(\mathbf{r})$, $\delta^{\text{RX-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$, $\delta^{\text{CEM-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$, $\delta^{\text{OSP-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\text{VSC}_j})$.

TABLE XIV

AUC VALUES CALCULATED FROM THE THREE 2-D ROC CURVES FOR HYDICE URBAN DETECTION RESULTS IN FIG. 18

detector	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{SNPR}	AUC_{TDBS}	AUC_{ODP}	Time (s)
$\delta^{\text{RX-AD}}(\mathbf{r})$	0.9872	0.2575	0.0361	1.2448	0.9511	7.1406	0.2215	1.2087	**0.346452**
$\delta^{\text{CRD}}(\mathbf{r})$	0.9641	0.3076	0.0136	1.2717	0.9504	22.5998	0.2939	1.2581	5.821708
$\delta^{\text{RX-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$	**0.9861**	0.6555	0.0192	1.6416	0.9668	34.1216	0.6363	1.6224	5.808477
$\delta^{\text{CEM-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$	0.9818	**0.6771**	0.0185	**1.6589**	0.9631	**36.5176**	**0.6586**	**1.6404**	5.686779
$\delta^{\text{OSP-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\text{VSC}_j})$	0.9842	0.2952	**0.0135**	1.2793	**0.9706**	21.9462	0.2817	1.2659	5.731333

From a target detection point of view, these airplanes can be visualized as known targets. So, this scene is more suitable for target detection rather than AD. So, the performance analysis concluded for the above two data scenes would be expected to be different. Once again, we also selected the overall best performance of each type i.e., $\delta^{\text{RX-AD}}(\mathbf{r})$, $\delta^{\text{CRD}}(\mathbf{r})$, $\delta^{\text{RX-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$, $\delta^{\text{CEM-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$, and $\delta^{\text{OSP-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\text{FSC}_j})$ where the only different detector is $\delta^{\text{OSP-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\text{FSC}_j})$ which is different $\delta^{\text{OSP-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\text{VSC}_j})$ in Figs. 17 and 18. Fig. 19(a)–(e) shows their detection maps. By visual inspection, it seemed that CRD produced the best result among all five types of ADs. However, the two ADs, , $\delta^{\text{CEM-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$ and $\delta^{\text{OSP-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\text{FSC}_j})$, also detected some target pixels at upper right corner which were not shown in the detection map of CRD, for example, a small horizontal stripe segment. These target pixels may be actually anomalous pixels but could be confirmed by ground truth. If they are indeed anomalies, then CRD would not be the best AD.

Table XIV further calculated the AUC values of the eight detection measures where the best results among these five ADs are boldfaced and their computing running times are also documented for reference. CRD was clearly the winner with significant differences in terms of TD and overall detection performance but did not perform well in BS compared with $\delta^{\text{CEM-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$ and $\delta^{\text{OSP-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\text{FSC}_j})$. Therefore, it is not surprised to see that CRD has been shown to outperform many existing ADs. However, if we include BS as a criterion to measure AD performance, this conclusion may be misleading.

A final remark is worth noting. Despite that both CEM-AD and OSP-AD did perform comparably with their best, two key features make them very different techniques. The first and foremost feature is that CEM-AD inverts the sample correlation matrix so as to achieve BS, whereas OSP-AD implements an OSP projection matrix to annihilate BKG. Because of their differences in functionality, CEM-AD and OSP-AD performed differently as illustrated in Tables XII–XV, specifically San Diego airport data. A primary reason that OSP-AD does not work well is mainly due to its use of OSP to annihilate the undesired signature matrix $\mathbf{U}$. If $\mathbf{U}$ is not properly determined, OSP-AD will not work as expected. More specifically, the

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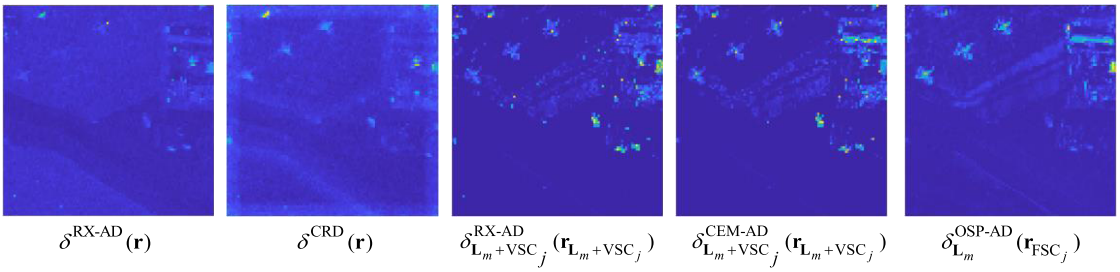


Fig. 19. Detection maps of five types of best representative ADs for San Diego airport, $\delta^{\text{RX-AD}}(\mathbf{r})$, $\delta^{\text{CRD}}(\mathbf{r})$, $\delta^{\text{RX-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$, $\delta^{\text{CEM-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$, $\delta^{\text{OSP-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\text{FSC}_j})$.

TABLE XV
AUC VALUES CALCULATED FROM THE THREE 2-D ROC CURVES FOR SAN DIEGO AIRPORT DETECTION RESULTS IN FIG. 19

detector	$\text{AUC}_{(D,F)}$	$\text{AUC}_{(D,\tau)}$	$\text{AUC}_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{SNPR}	AUC_{TDBS}	AUC_{ODP}	Time (s)
$\delta^{\text{RX-AD}}(\mathbf{r})$	0.8334	0.0910	0.0453	0.9244	0.7861	2.0101	0.0457	0.8791	**0.364277**
$\delta^{\text{CRD}}(\mathbf{r})$	**0.9964**	**0.4030**	0.0438	**1.3994**	**0.9524**	9.1900	**0.3591**	**1.3556**	5.424751
$\delta^{\text{RX-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$	0.9336	0.1332	**0.0181**	1.0668	0.9154	7.3383	0.1150	1.0487	8.165311
$\delta^{\text{CEM-AD}}_{\mathbf{L}_m+\text{VSC}_j}(\mathbf{r}_{\mathbf{L}_m+\text{VSC}_j})$	0.9347	0.1809	0.0188	1.1156	0.9158	9.6212	0.1621	1.0968	8.046591
$\delta^{\text{OSP-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\text{FSC}_j})$	0.9600	0.2785	0.0282	1.2385	0.9317	**9.8857**	0.2503	1.2103	12.233089

effectiveness of OSP-AD has nothing to do with OSP-AD itself but rather how to select an appropriate **U** for annihilation. For more details of experiments and discussions on these two approaches, we refer [11] and [33].

X. CONCLUSION

HTD and HAD have been shown to be dual theories under an umbrella of SNR [1]. This article develops a DVT to confirm this solid evidence. In particular, it reinterprets the minimum variance resulting from CEM as maximum SBR. With imposing SBR of a UST as a constraint on CEM instead of unity imposed on the known target signature as CEM does, we can derive an SBR-CEM which further yields UST-CEM. By virtue of UST-CEM, DVT is then developed and used to derive a CEM-AD which shows that the well-known R-AD developed in [3], [8], [31] is indeed a special case of CEM-AD. Furthermore, a similar treatment of using UST as a dummy variable by DVT for CEM can also be used to further show that the OSP-AD developed in [11] can be rederived by UST-OSP via DVT. So, one main contribution of this article is to develop SBR-CEM which yields DVT. This further confirms that HAD is in fact a dual theory of HTD [1] because RX-AD, CEM-AD, and OSP-AD can actually be derived by GLRT, CEM, and OSP via DVT under SNR or SBR criterion, respectively. It is our brief that this connection is very significant and provides additional insights into HAD. A second major contribution is to introduce two types of SC, FSC and VSC, to constrain anomalies in the sparse space **S** generated by GoDec so that anomalies can be separated from noise in **S**. A third contribution is that CEM-AD operates the data sample **r** and the correlation matrix **R** in two individual separate spaces. This is quite different from traditional ADs which operate **r** and **R** in the same data space. Finally, to better evaluate AD performance, 3-D ROC curve-derived detection measures are further used to measure TD, BS, joint TD and BS, and overall detection performance.

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Chein-I Chang (Life Fellow, IEEE) received the B.S. degree in mathematics from Soochow University, Taipei, Taiwan, in 1973, the M.S. degree in mathematics from the Institute of Mathematics, National Tsing Hua University, Hsinchu, Taiwan, in 1975, the M.A. degree in mathematics from the State University of New York, Stony Brook, NY, USA, in 1977, the M.S. and M.S.E.E. degrees from the University of Illinois at Urbana–Champaign, Urbana, IL, USA, in 1982, and the Ph.D. degree in electrical engineering from the University of Maryland, College Park, MD, USA, in 1987.

He has been with the University of Maryland, Baltimore County (UMBC) since 1987, where he is a Professor with the Department of Computer Science and Electrical Engineering. He authored four books, *Hyperspectral Imaging: Techniques for Spectral Detection and Classification* published by Kluwer Academic Publishers in 2003, *Hyperspectral Data Processing: Algorithm Design and Analysis* by John Wiley & Sons, 2013, *Real Time Progressive Hyperspectral Image Processing*: *Endmember Finding and Anomaly Detection* 2016 by Springer, and *Recursive Hyperspectral Sample and Band Processing: Algorithm Architecture and Implementation* by Springer 2017. In addition, he also edited two books, *Recent Advances in Hyperspectral Signal and Image Processing*, 2006 and *Hyperspectral Data Exploitation: Theory and Applications*, John Wiley & Sons, 2007, and co-edited with A. Plaza a book, *High Performance Computing in Remote Sensing*, CRC Press, 2007. He is also editing a book, *Advances in Hyperspectral Image Processing Techniques*, to be published by Wiley, 2022. He currently serves as an Associate Editor for *Remote Sensing* and IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING. His research interests include multispectral/hyperspectral image processing, automatic target recognition, and medical imaging.

Dr. Chang is a fellow of SPIE. He was a recipient of the National Research Council Senior Research Associateship Award from 2002 to 2003 sponsored by U.S. Army Soldier and Biological Chemical Command, Edgewood Chemical and Biological Center, Aberdeen Proving Ground, Maryland.

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